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Espinosa

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(54) **HOLD DOWN SYSTEM AND BUILDING USING THE SAME**

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(60) Provisional application No. 60/843,698, filed on Sep. 12, 2006.

(51) **Int. Cl.**
E04H 9/14 (2006.01)
E02D 27/34 (2006.01)
E04B 1/26 (2006.01)
E04B 1/41 (2006.01)
E04B 2/70 (2006.01)
E04C 5/16 (2006.01)
F16B 43/00 (2006.01)
E04B 1/35 (2006.01)

(52) **U.S. Cl.**

CPC **E04B 1/4157** (2013.01); **E02D 27/34** (2013.01); **E04B 1/2604** (2013.01); **E04B 2/70** (2013.01); **E04H 9/14** (2013.01); **F16B 43/00** (2013.01); **E04B 2001/2688** (2013.01); **E04B 2001/3583** (2013.01); **E04C 5/16** (2013.01); **Y02A 50/00** (2018.01)

(58) **Field of Classification Search**

None
See application file for complete search history.

(56)

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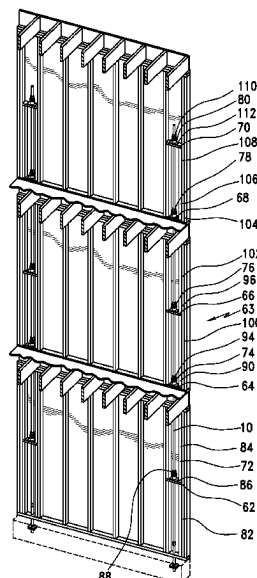
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(57)

ABSTRACT

A building includes a stud wall including a first bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the first bottom plate within the stud wall; a first bridge member disposed on the first bottom plate, the first bridge member having an opening through which the tie rod extends; a pair of spaced apart reinforcement studs; and the pair of spaced apart reinforcement studs including bottom ends bearing on the first bridge member.

9 Claims, 24 Drawing Sheets

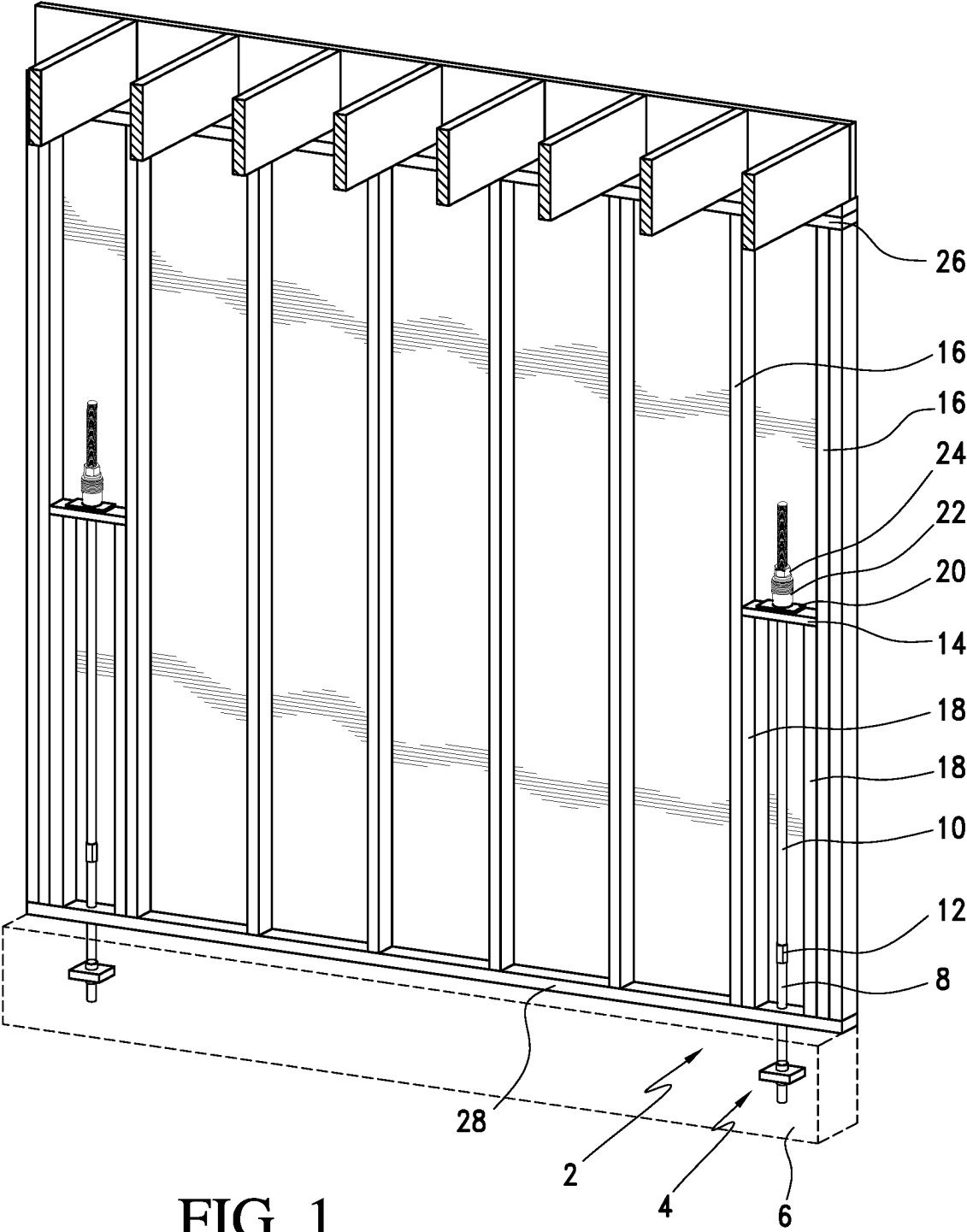


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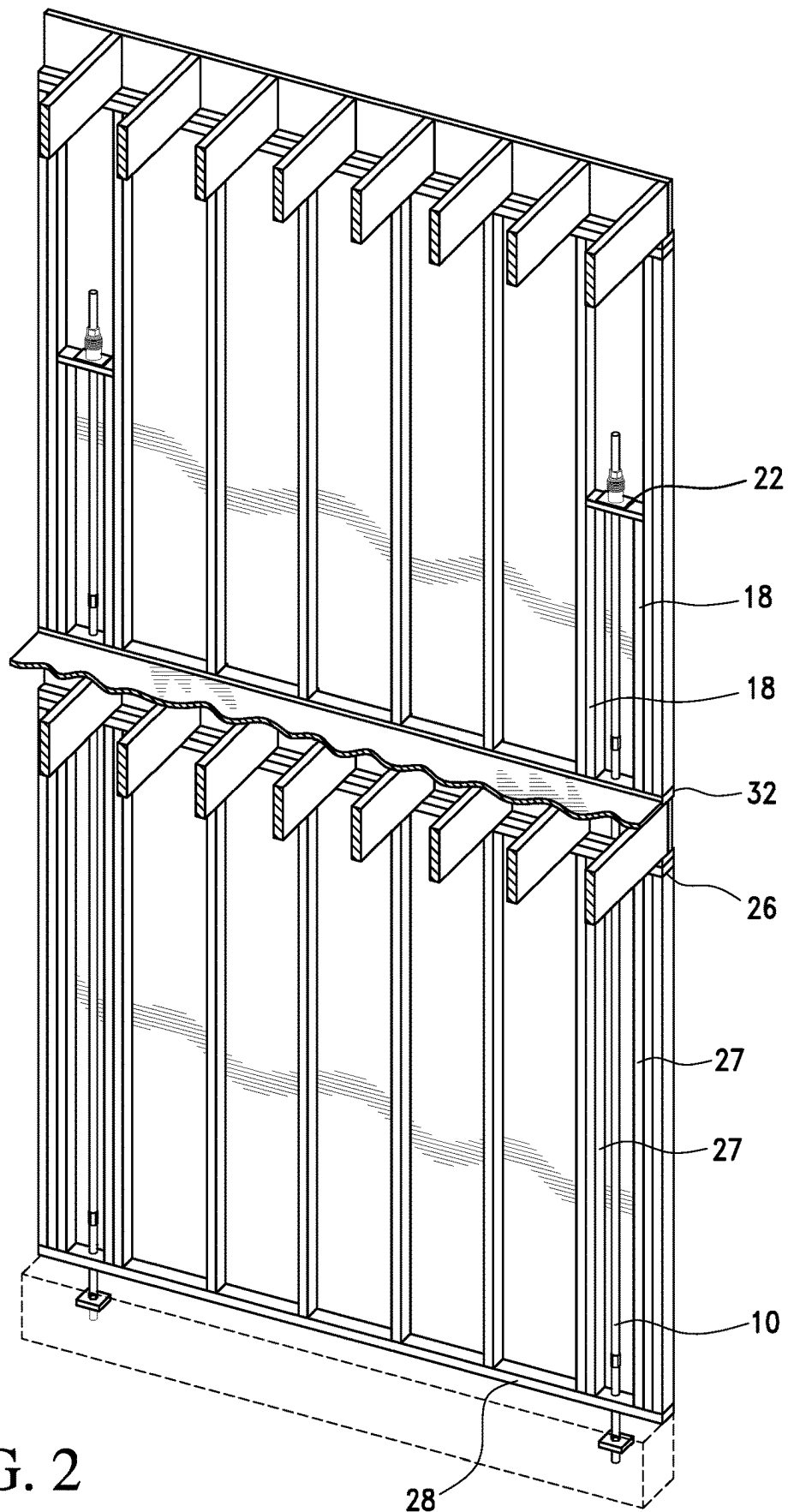


FIG. 2

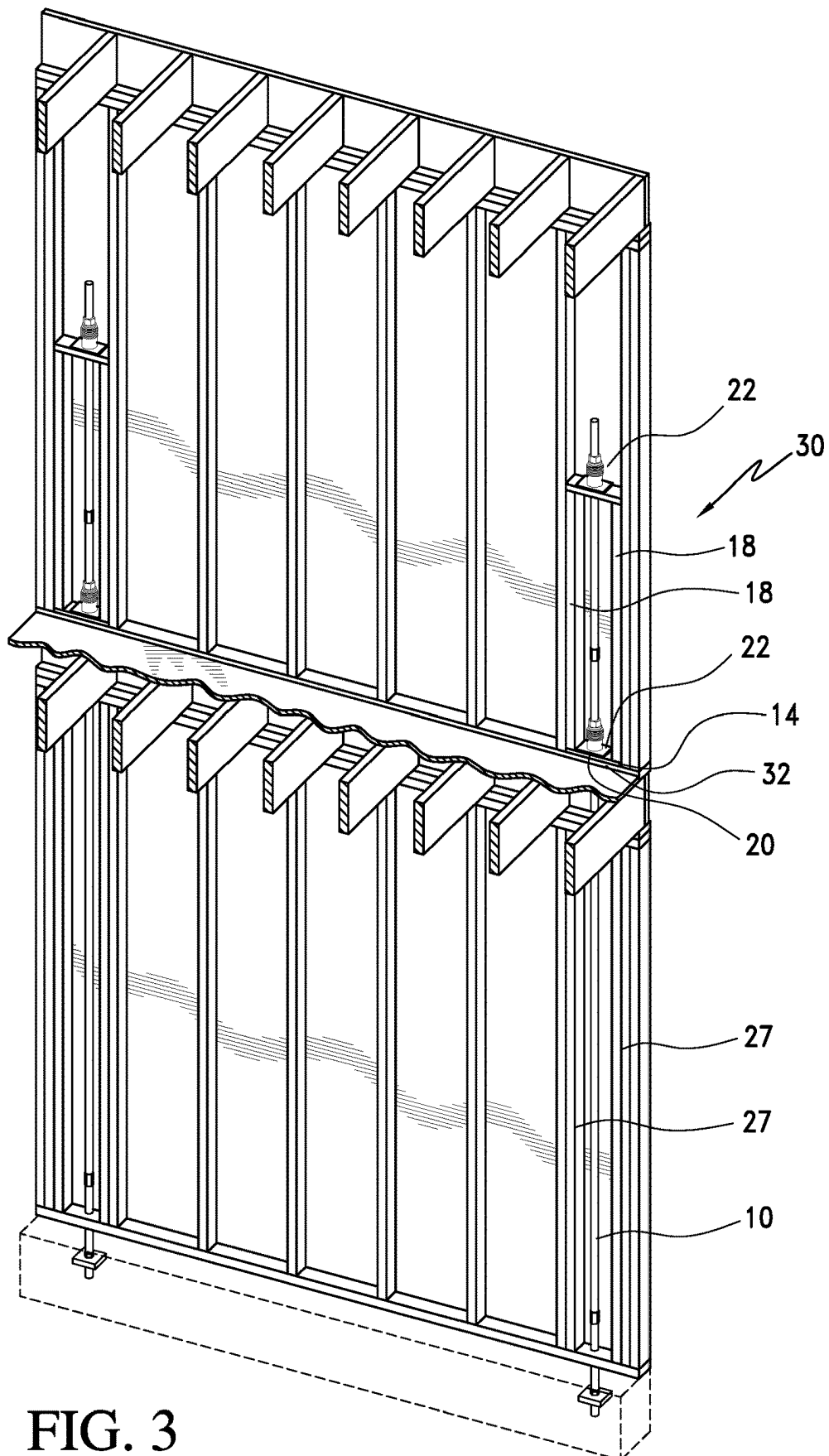


FIG. 3

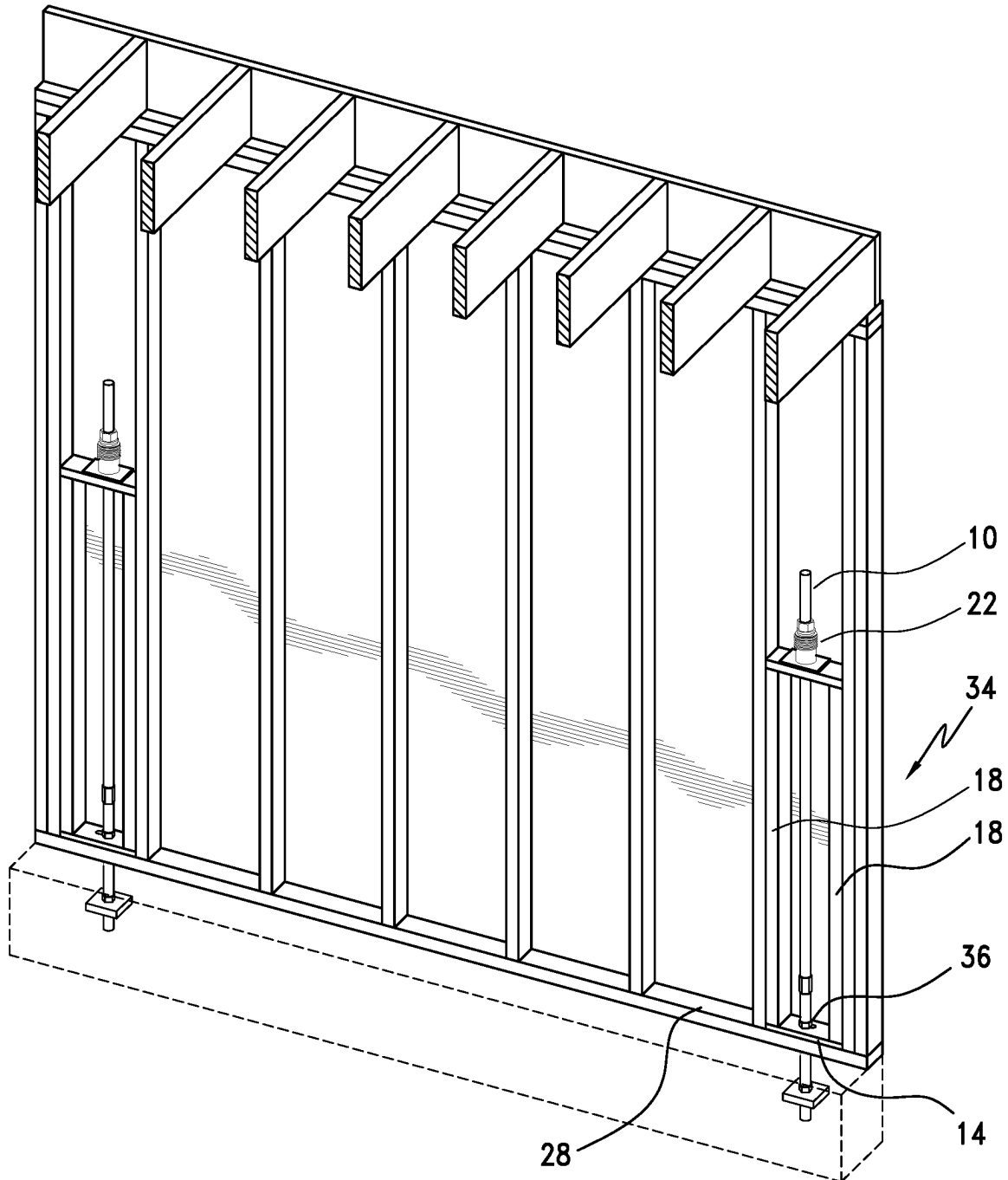


FIG. 4

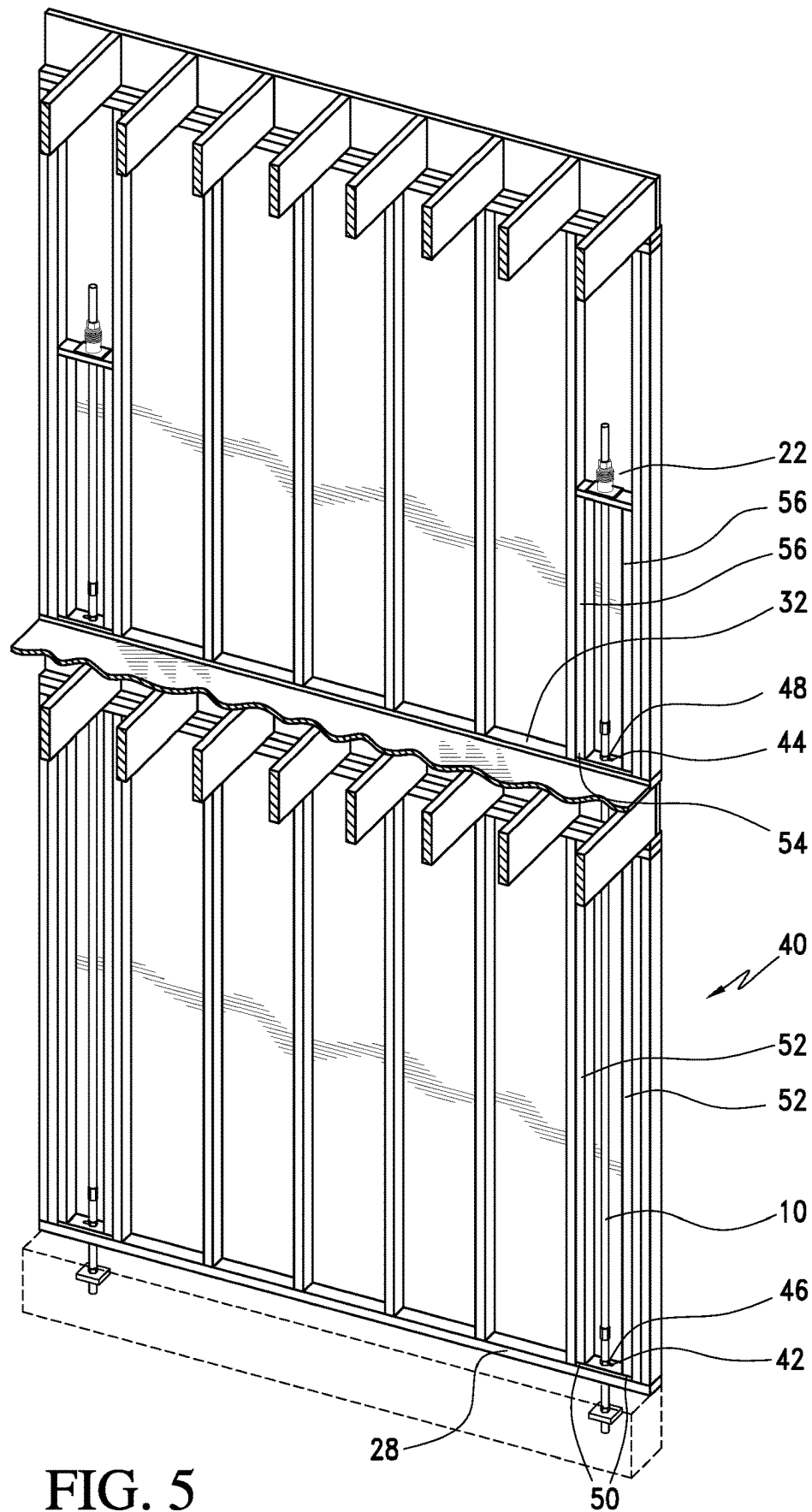
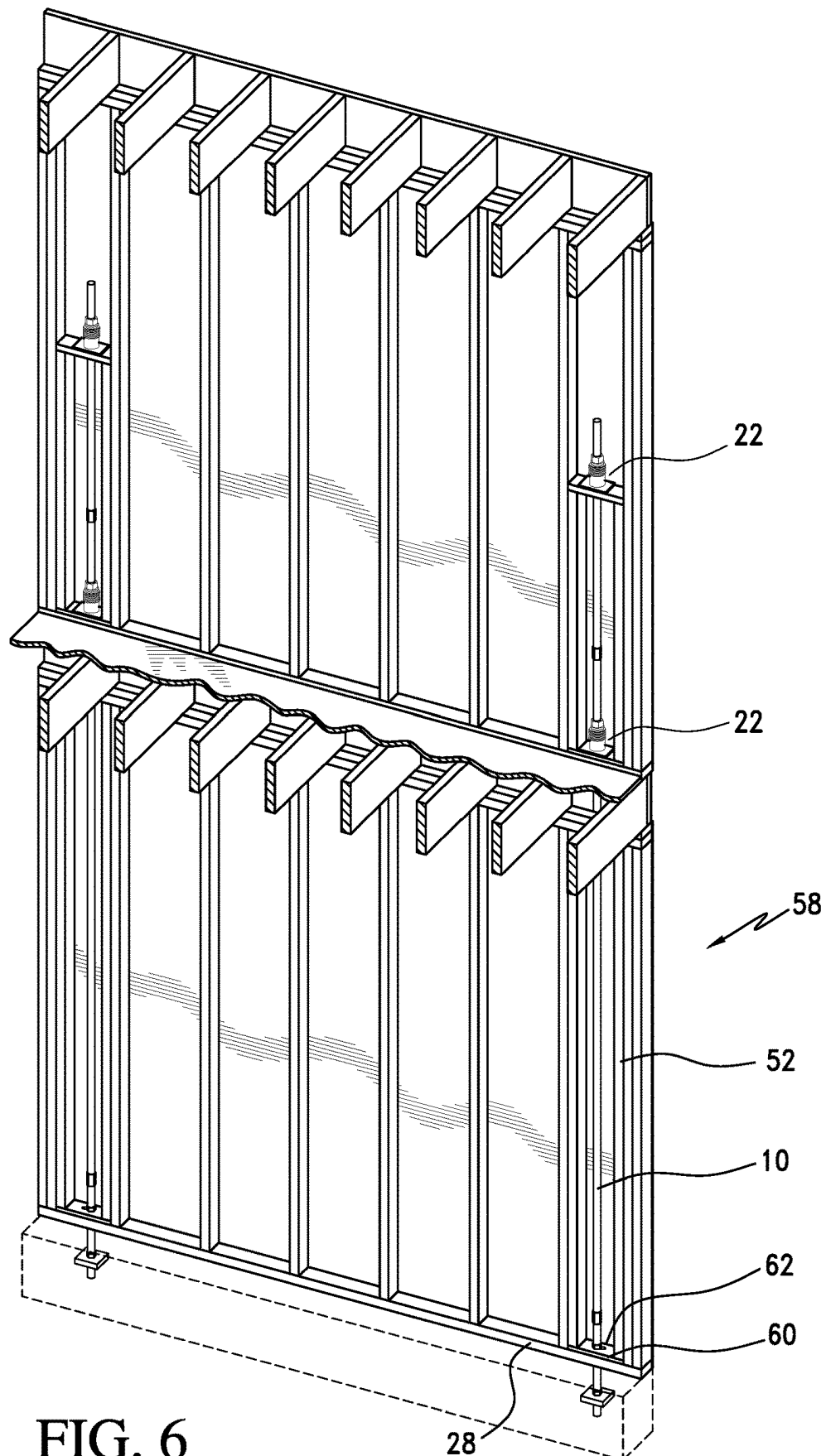


FIG. 5



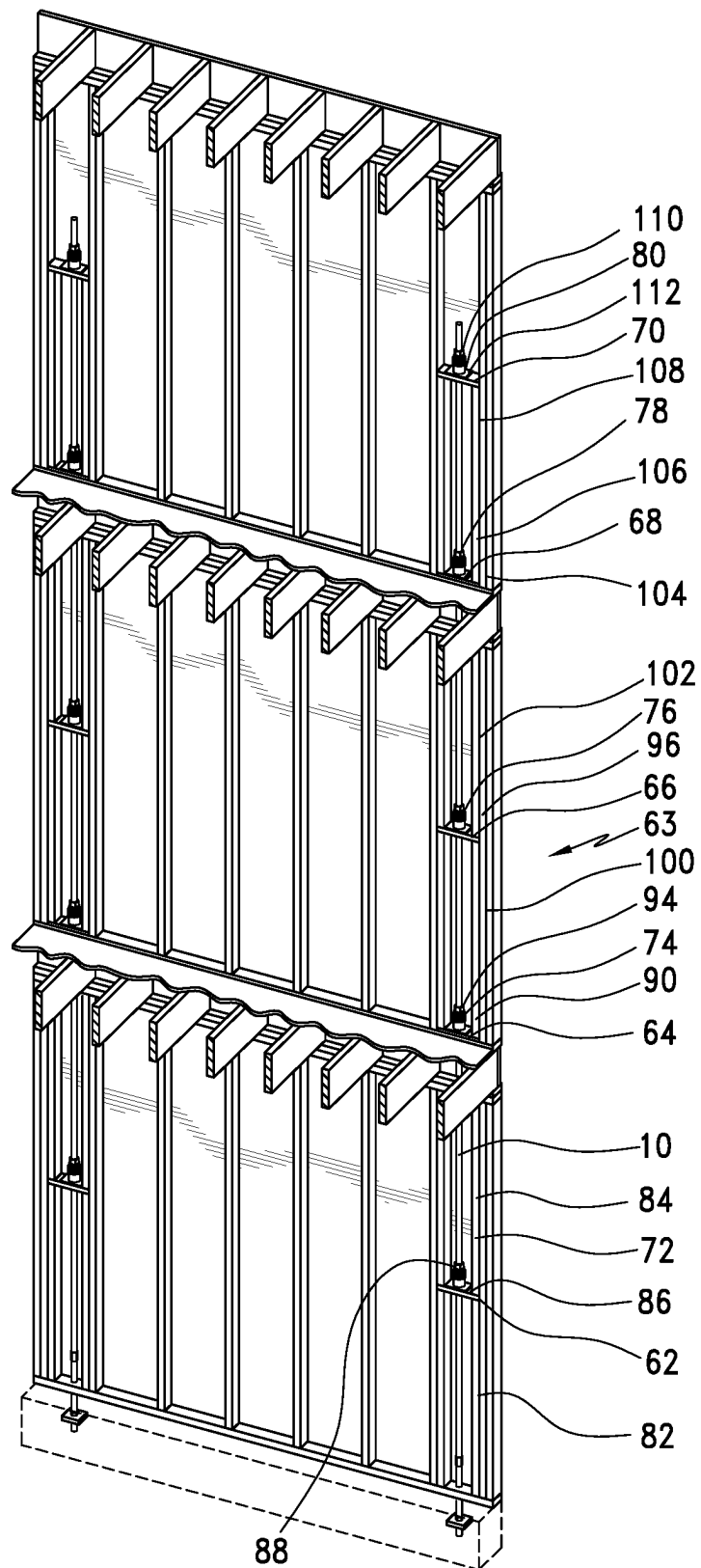


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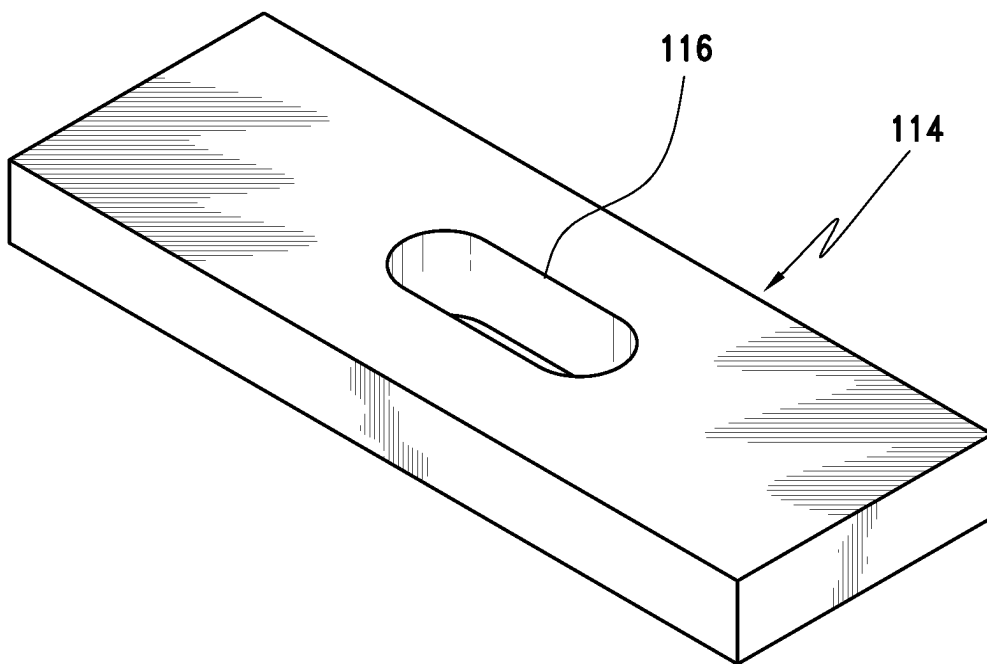


FIG. 8

FIG. 9

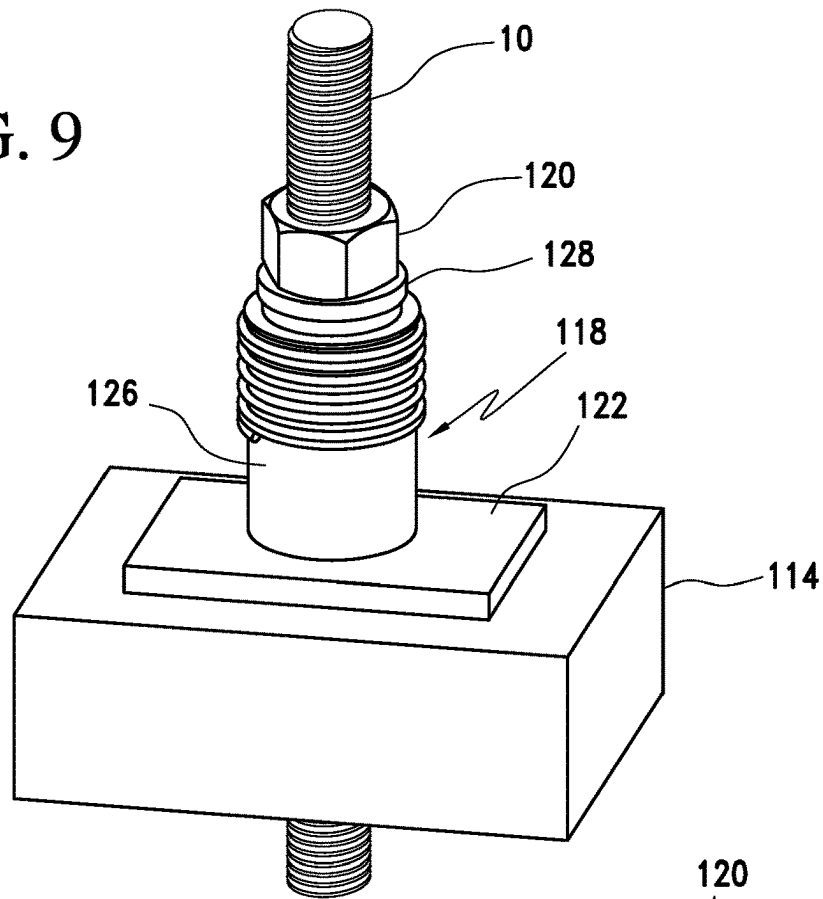


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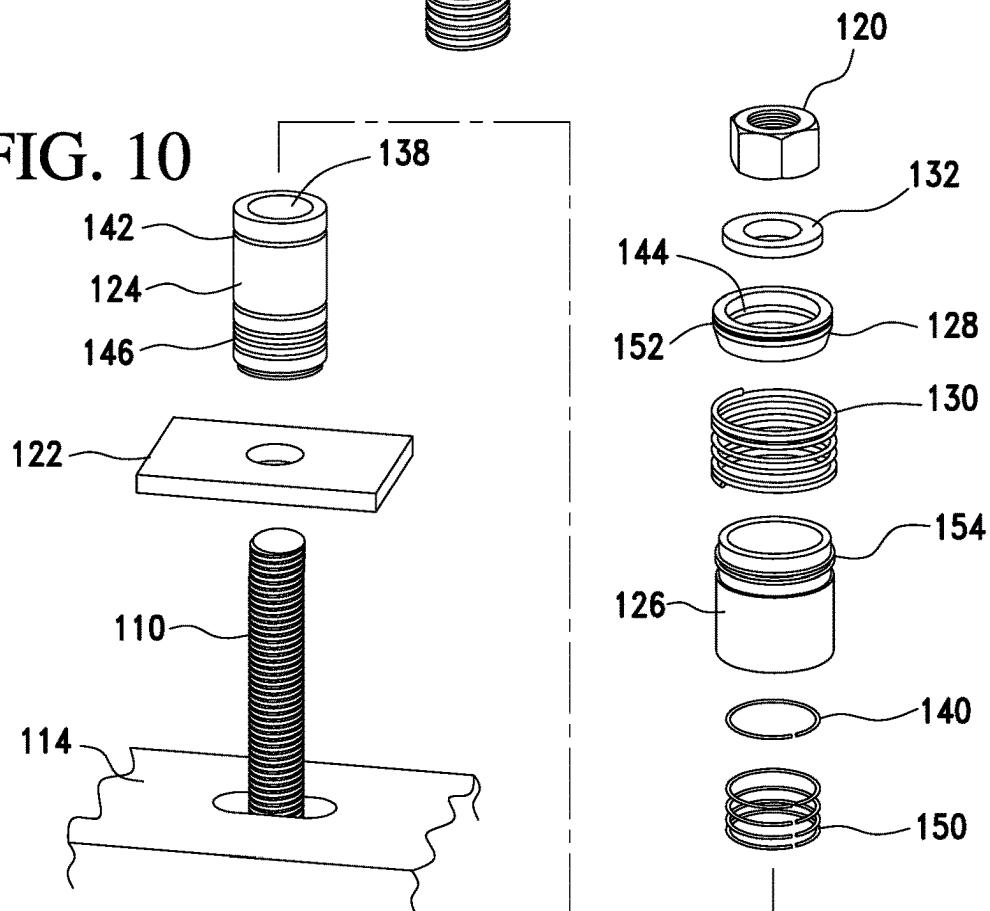


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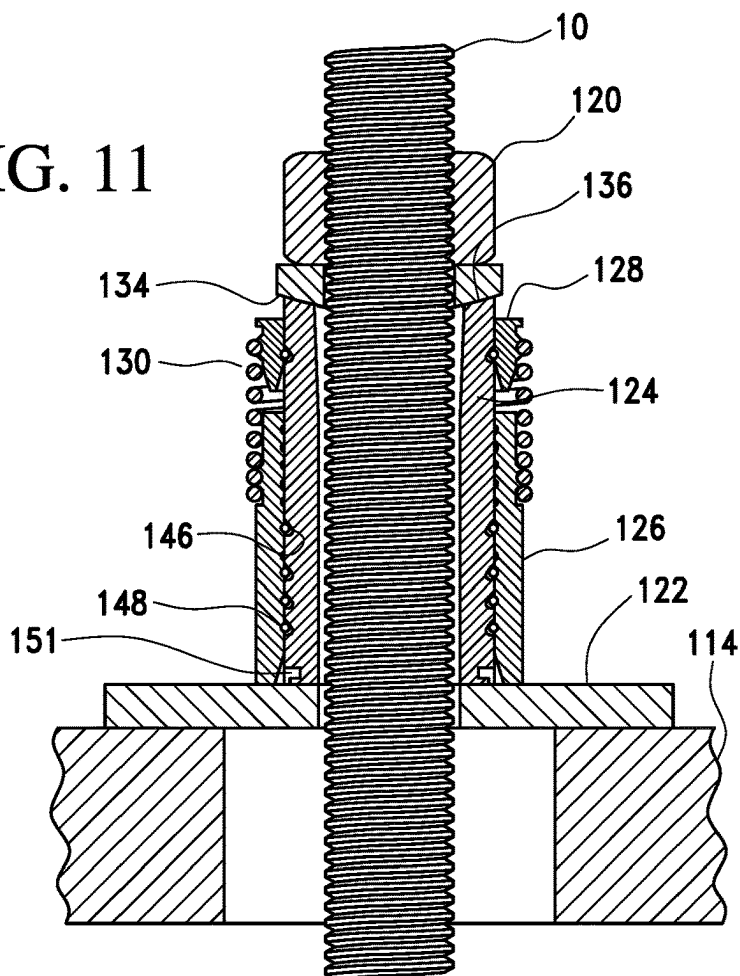
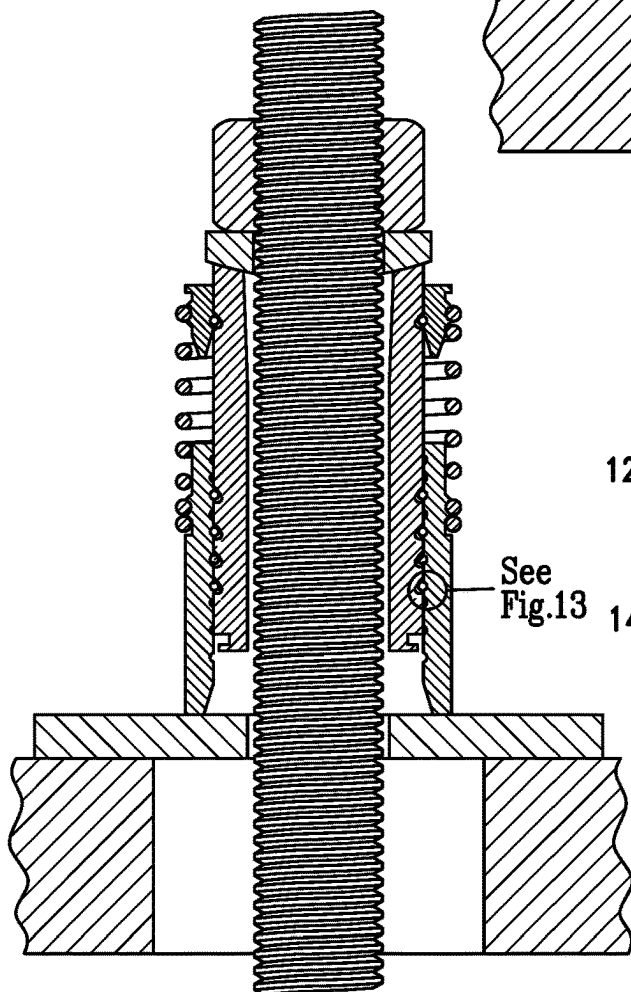


FIG. 12



See
Fig.13

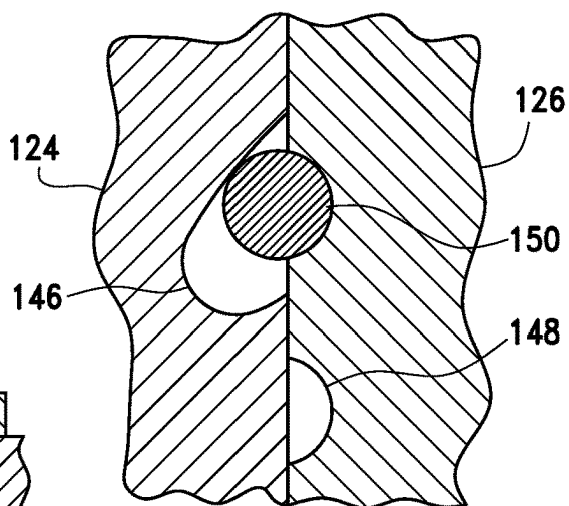


FIG. 13

FIG. 14

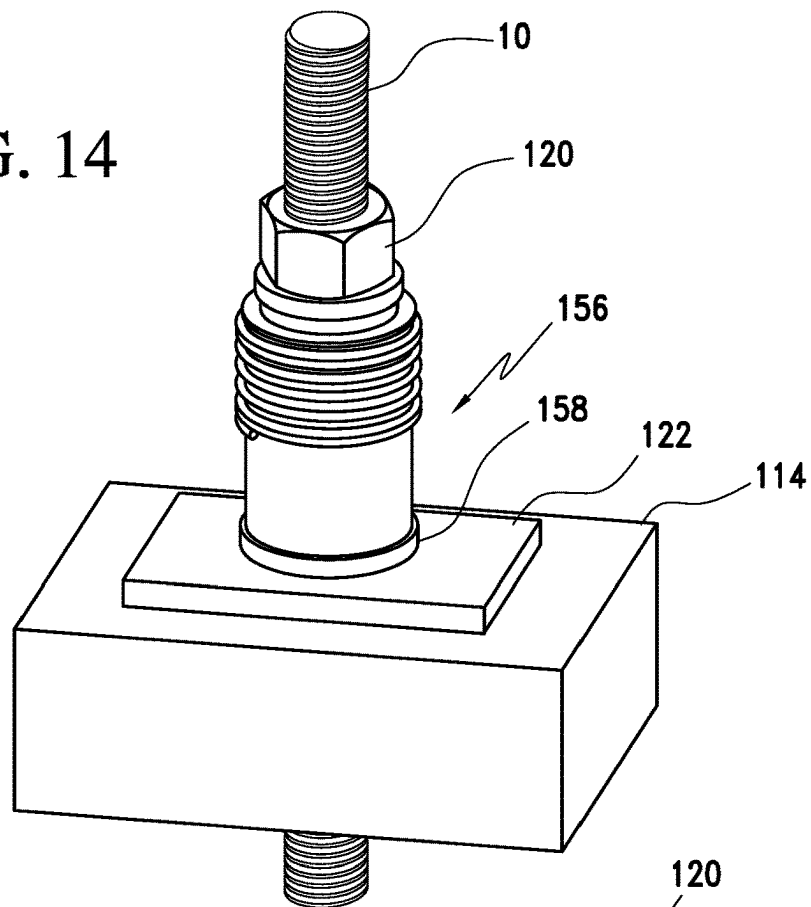


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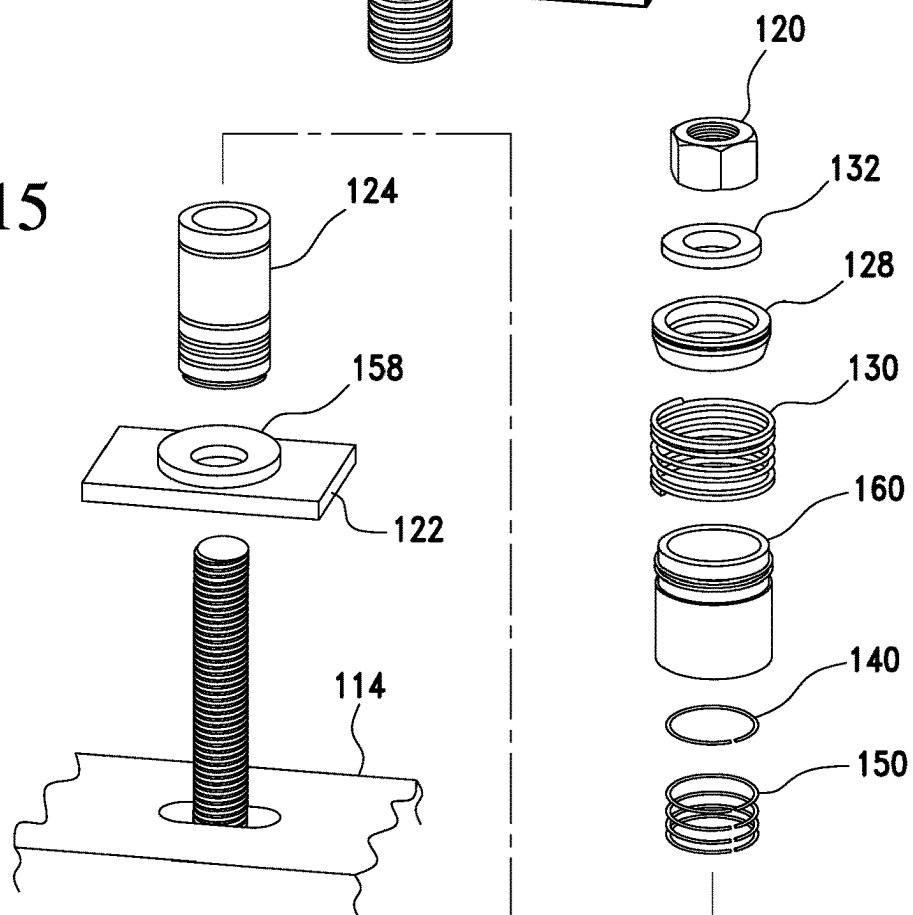


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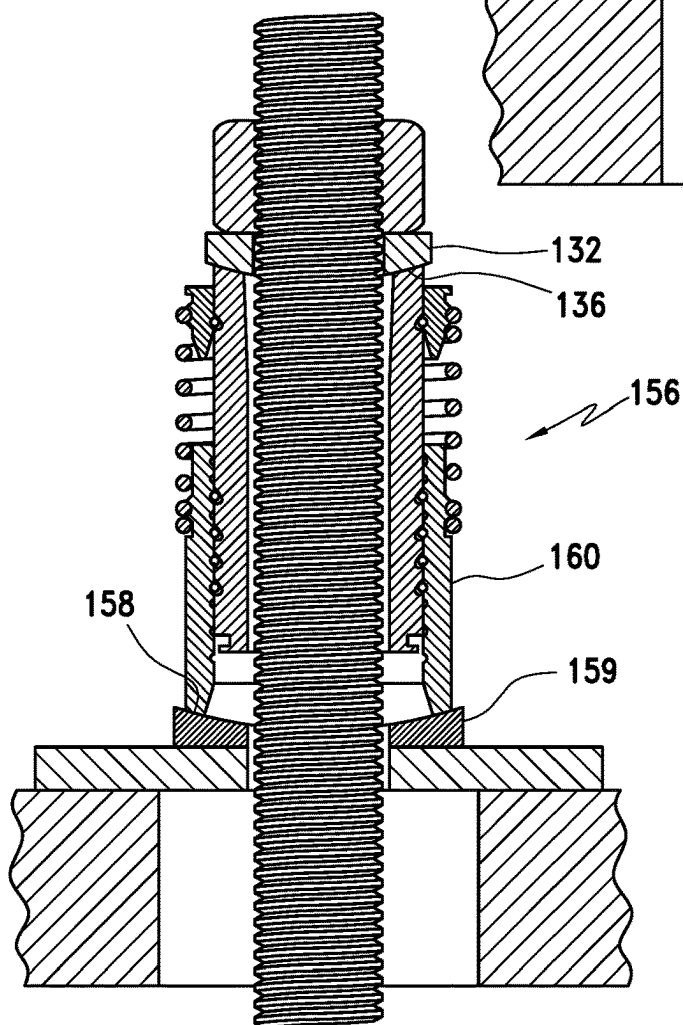
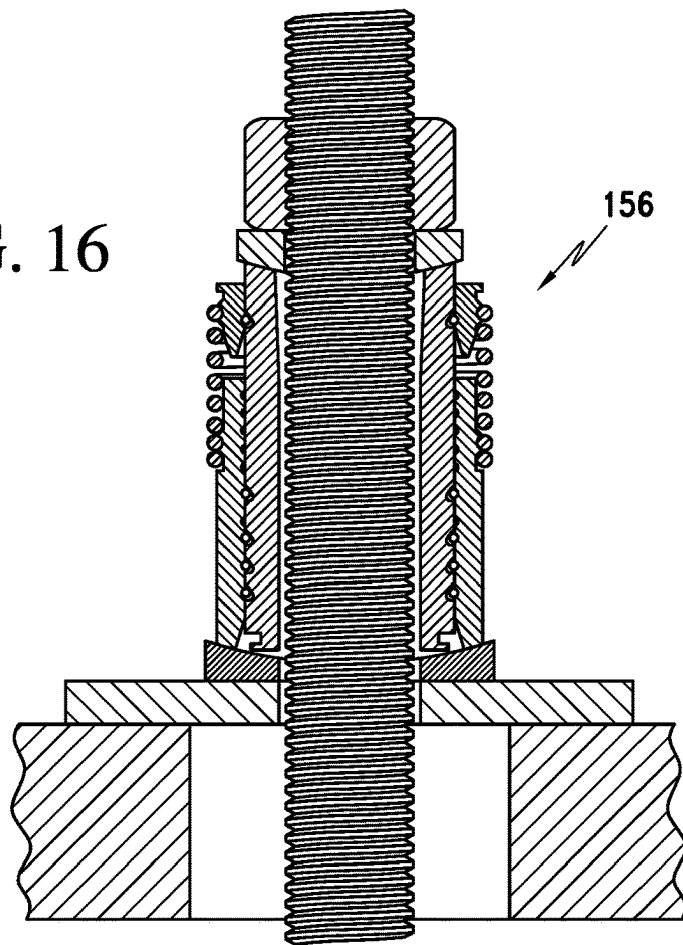


FIG. 17

FIG. 18

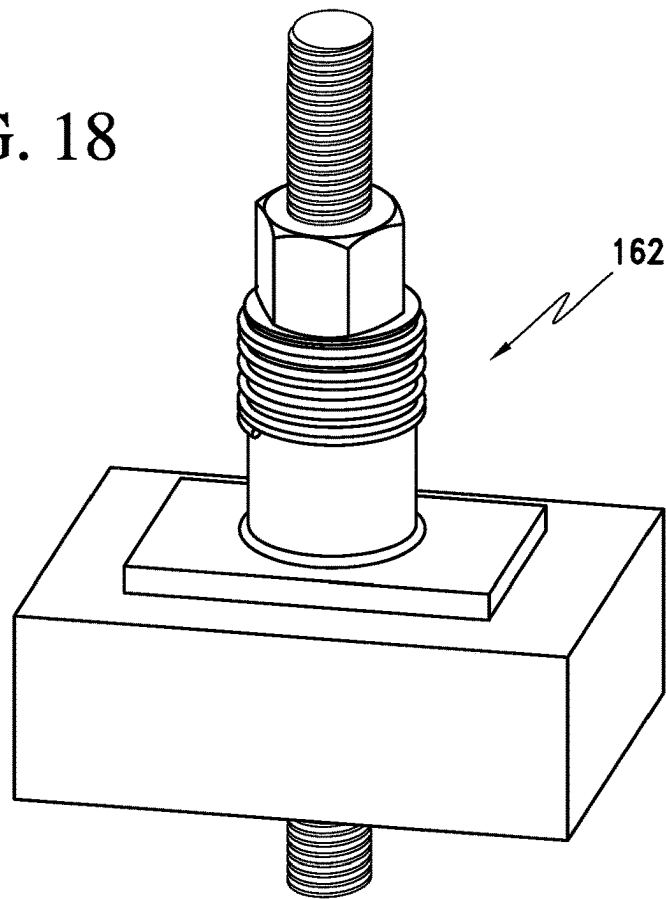


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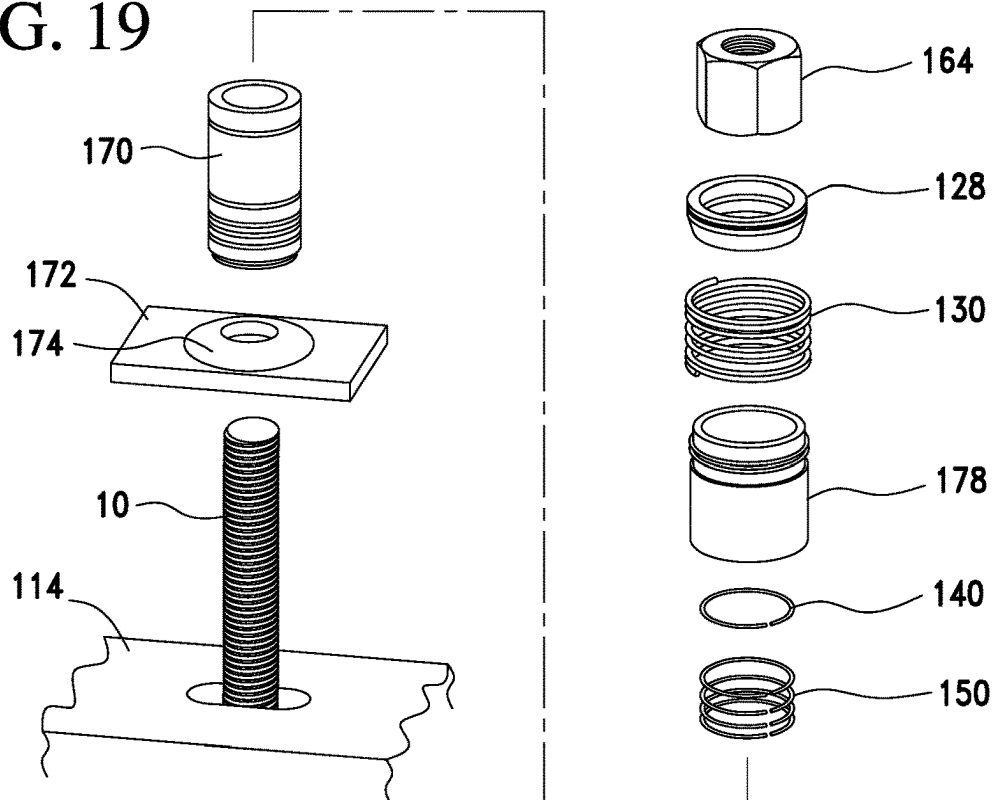


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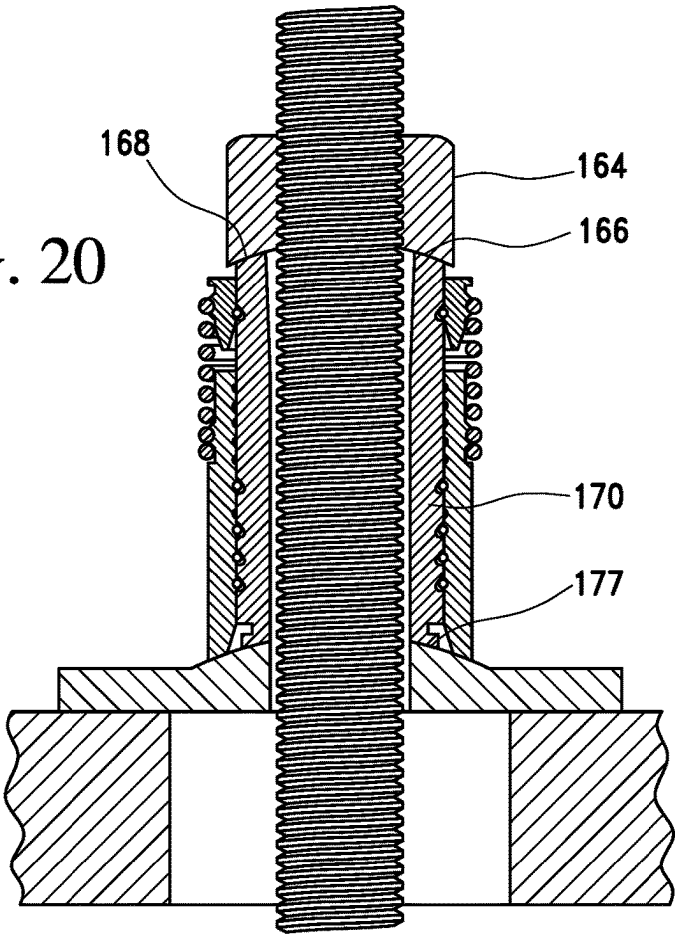


FIG. 21

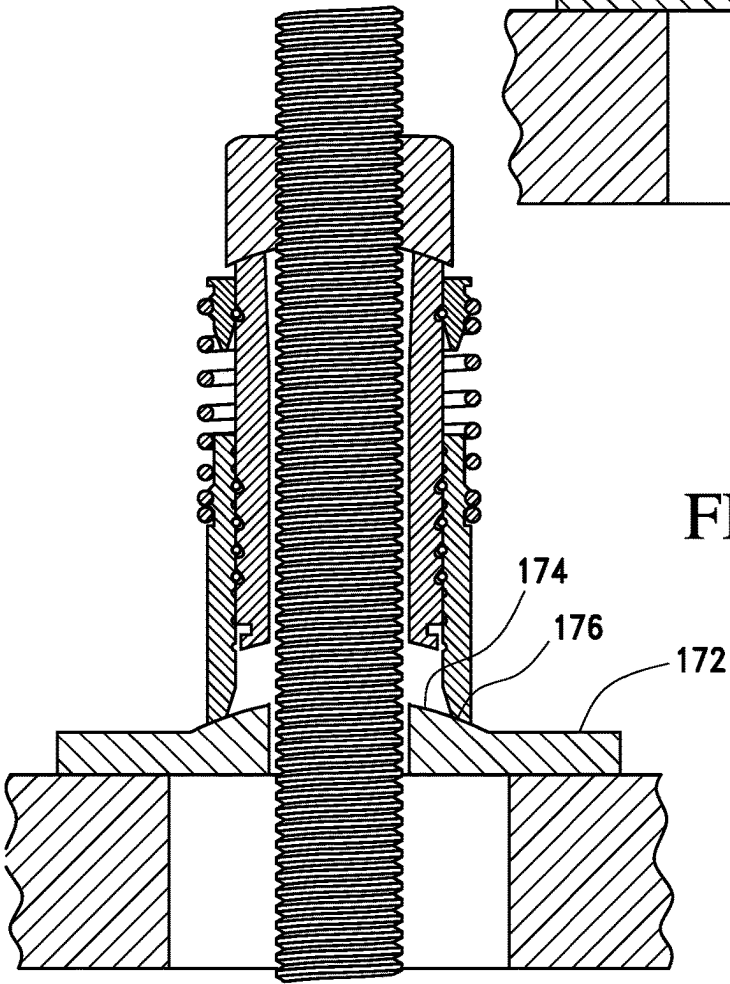


FIG. 22

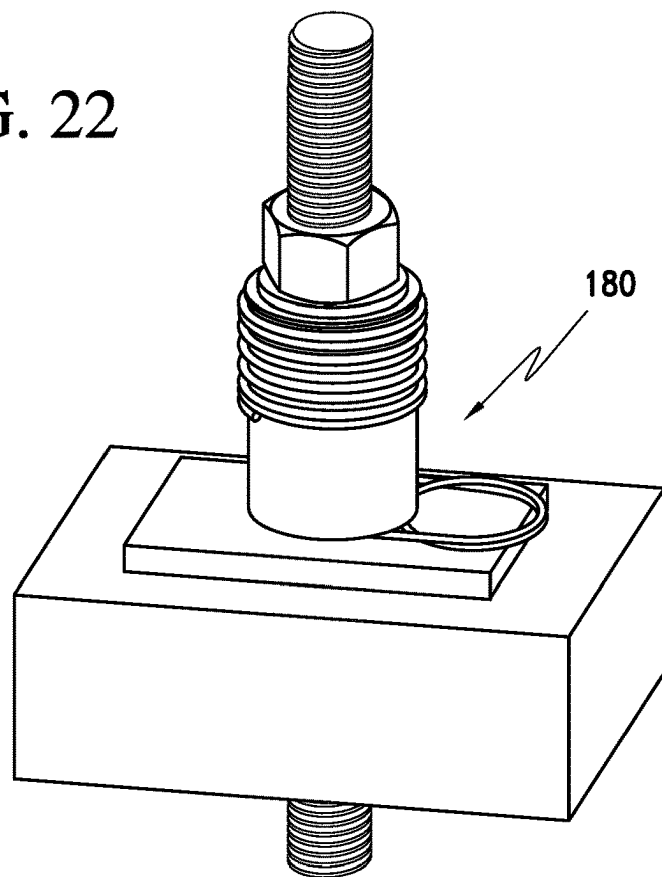


FIG. 23

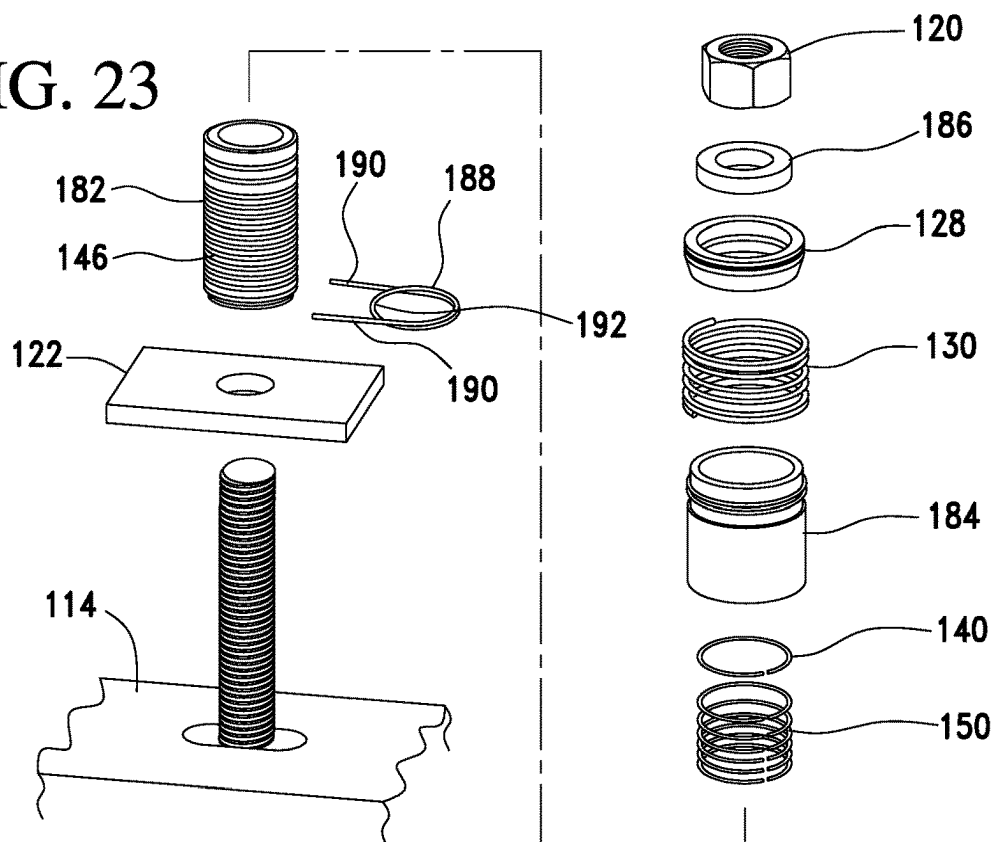


FIG. 24

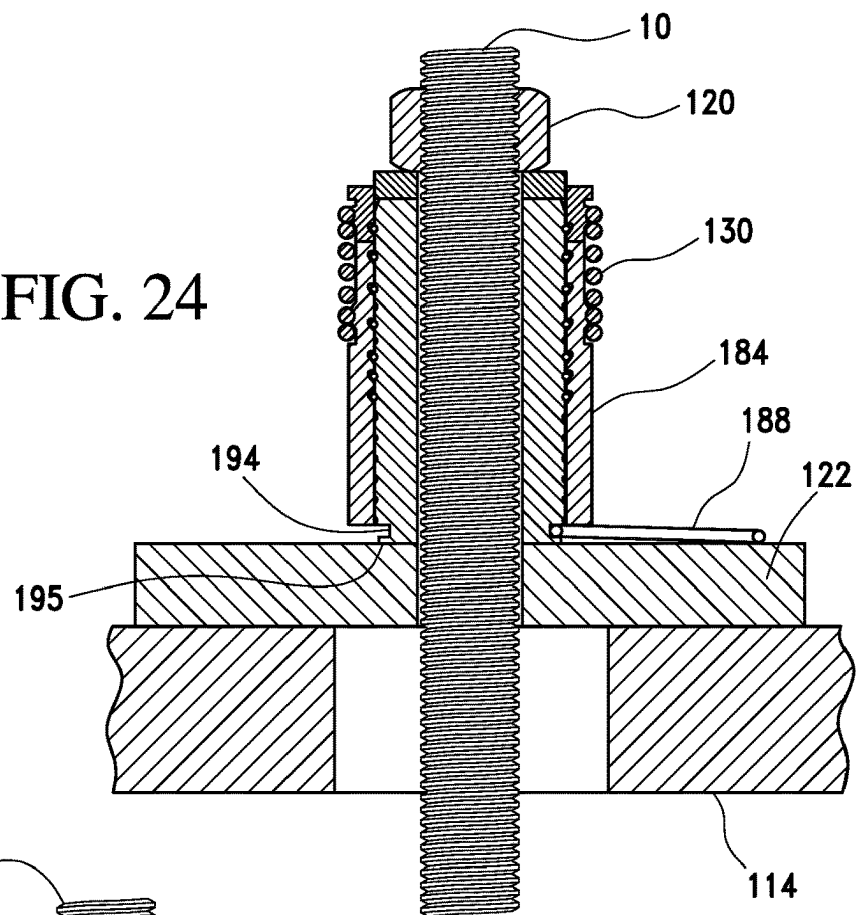


FIG. 25

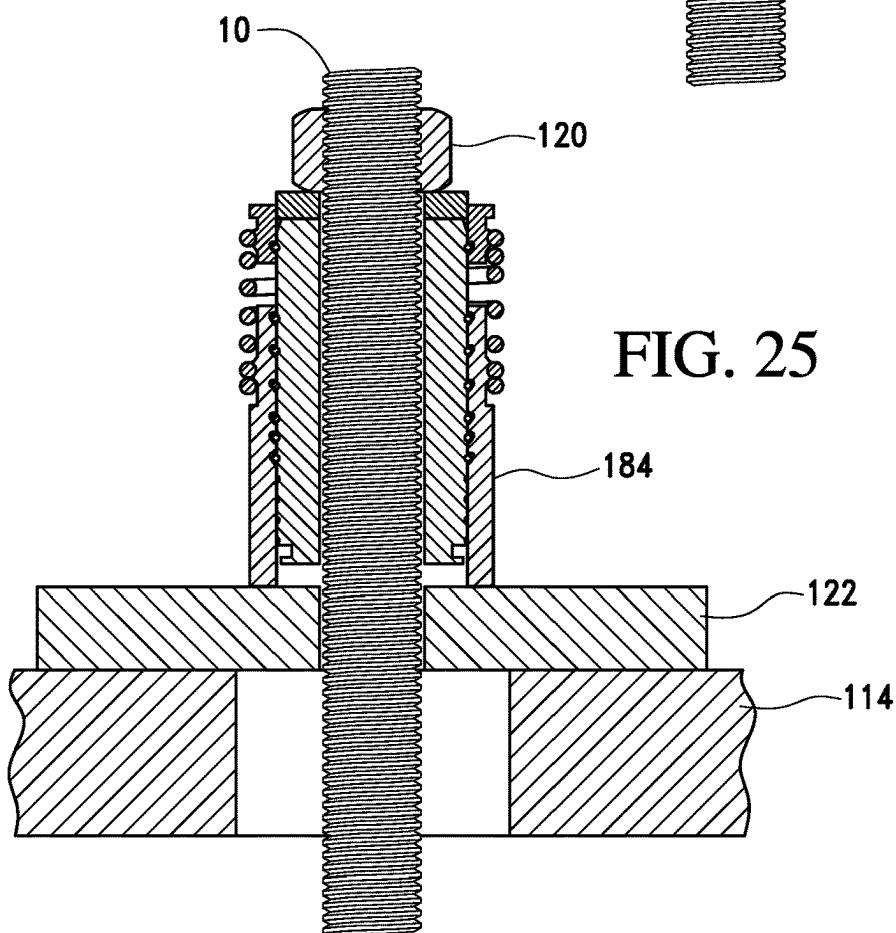


FIG. 26

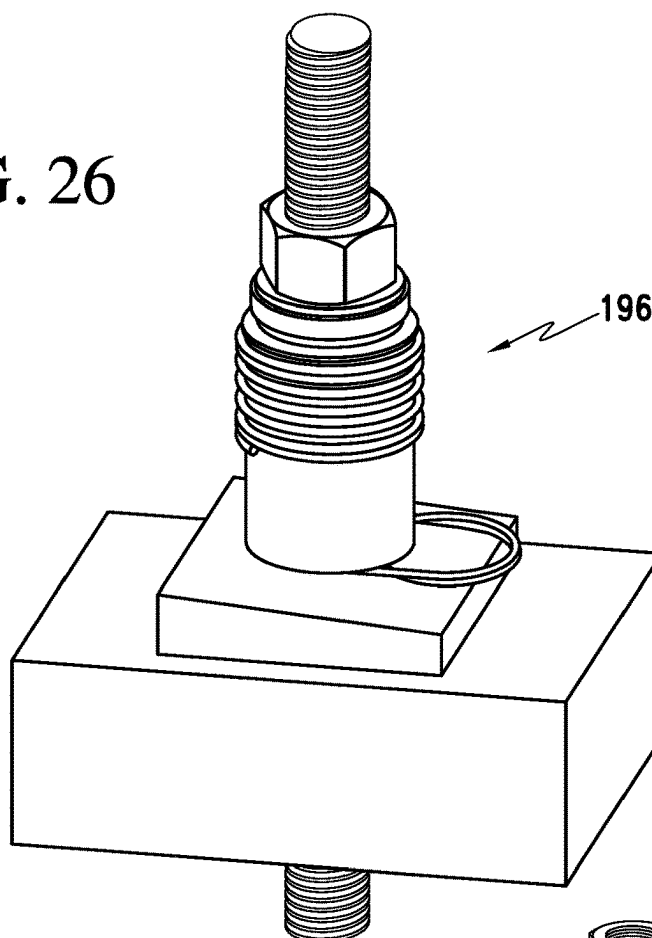
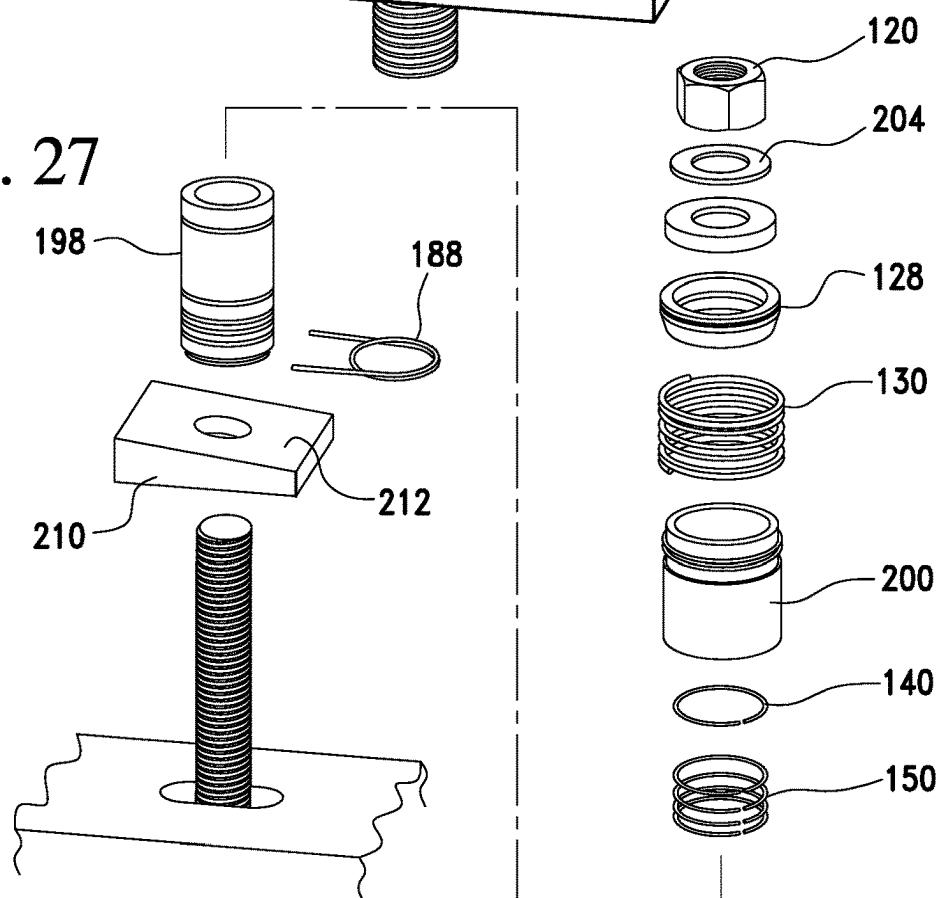


FIG. 27



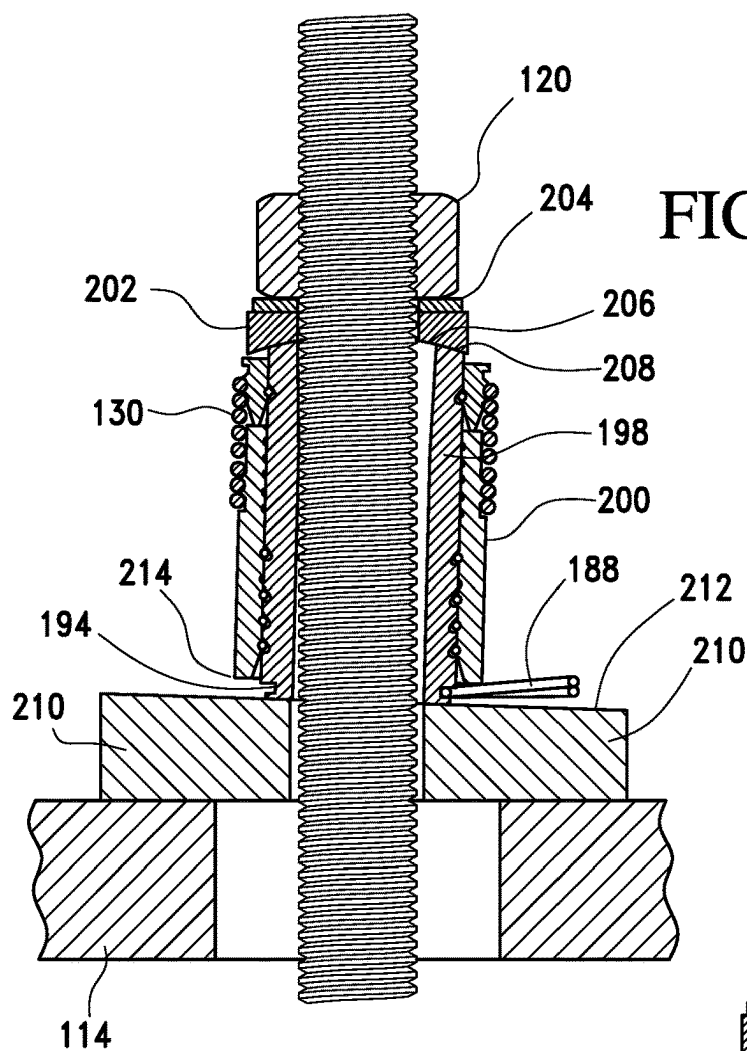


FIG. 29

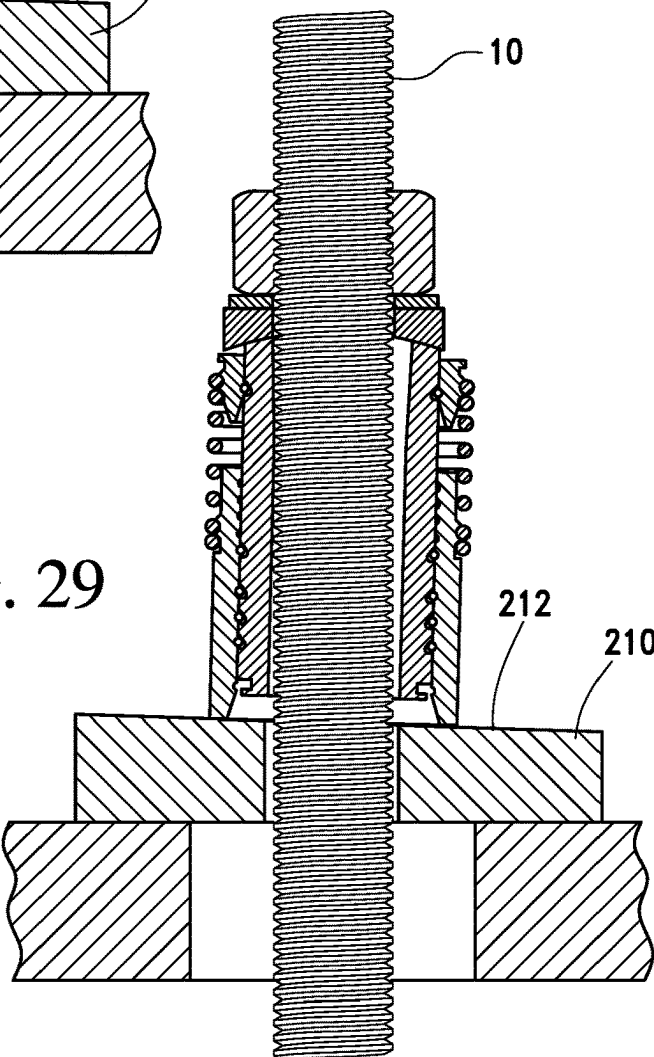


FIG. 30

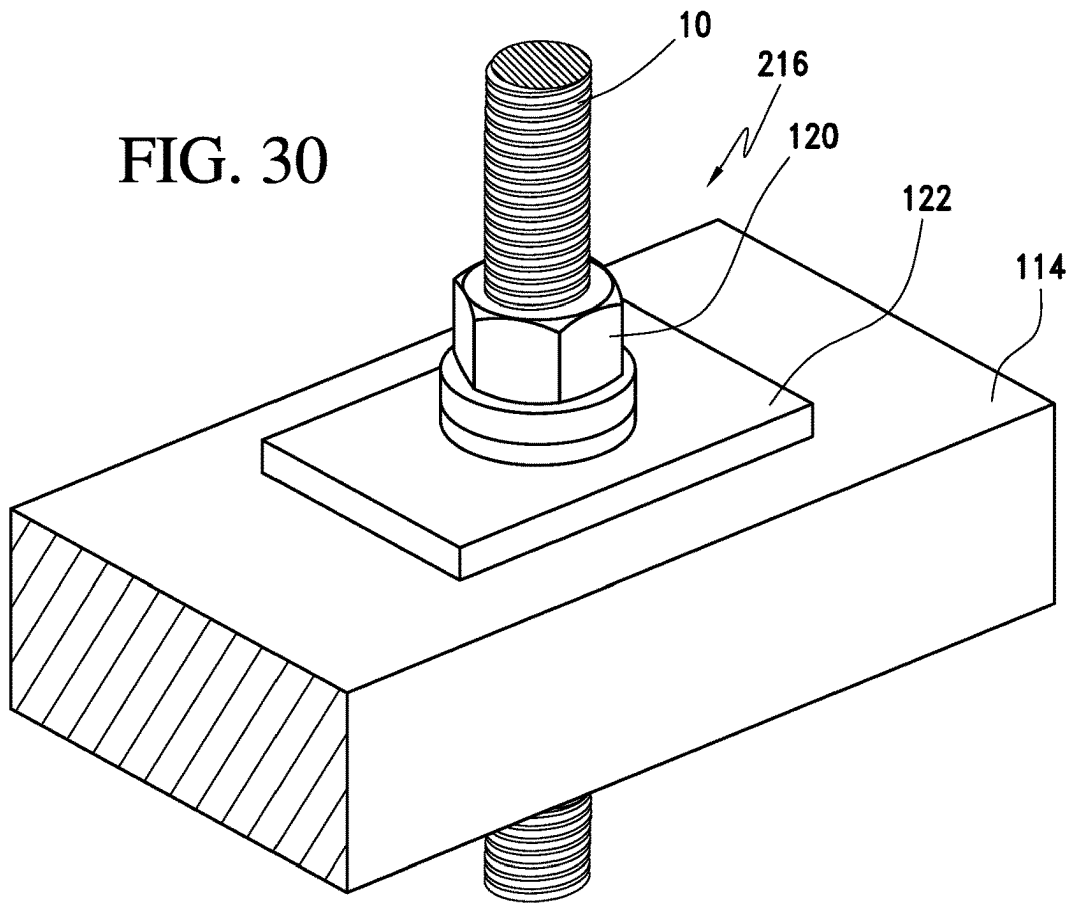
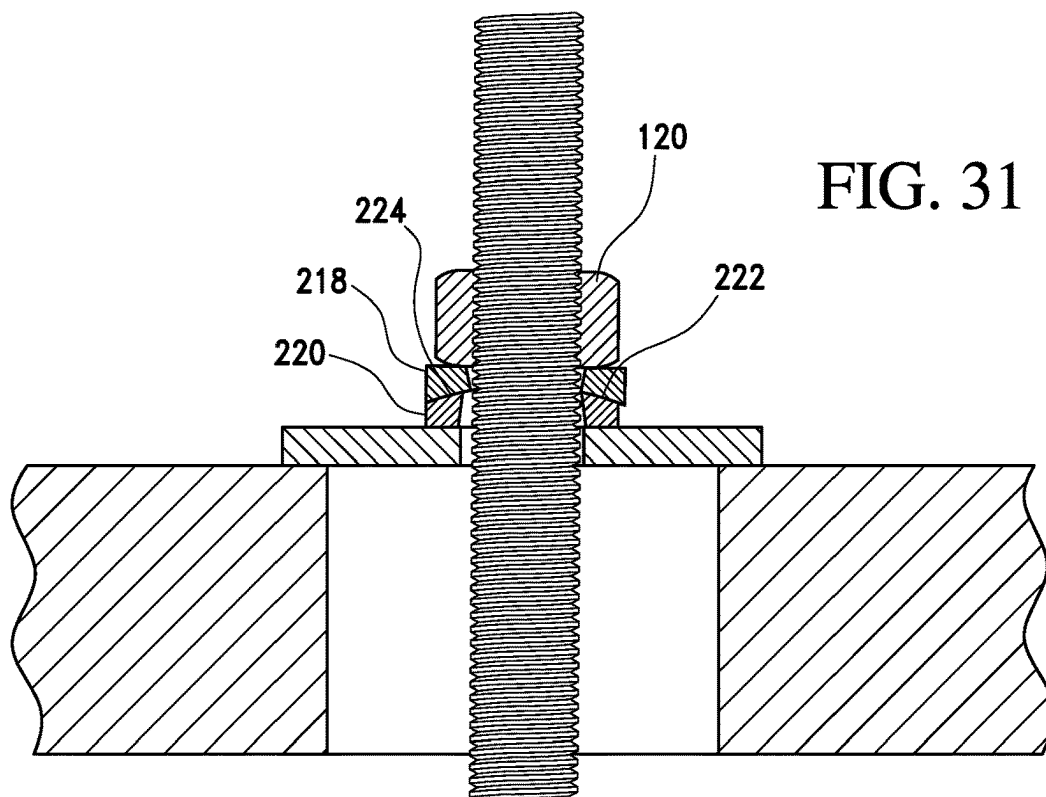


FIG. 31



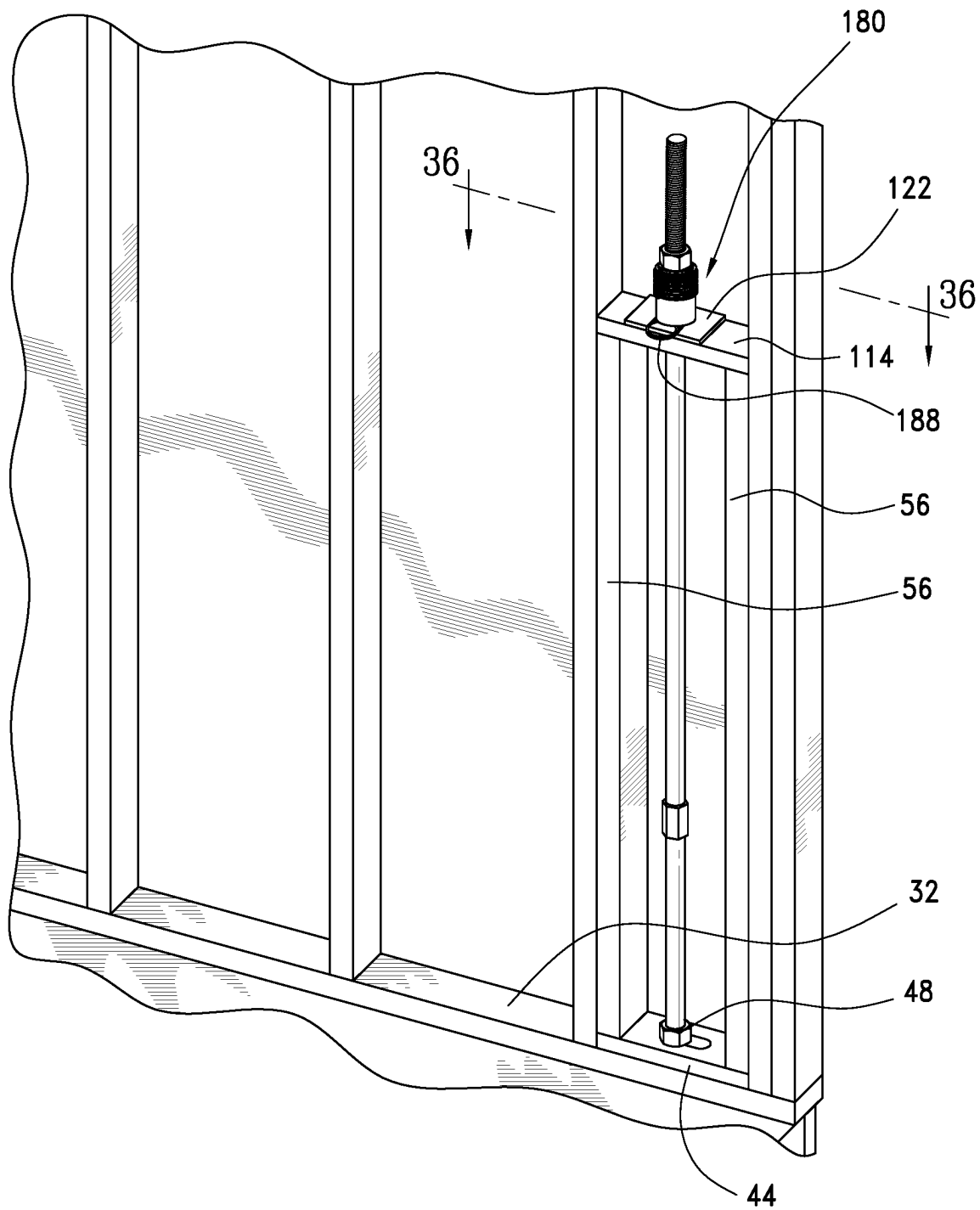


FIG. 32

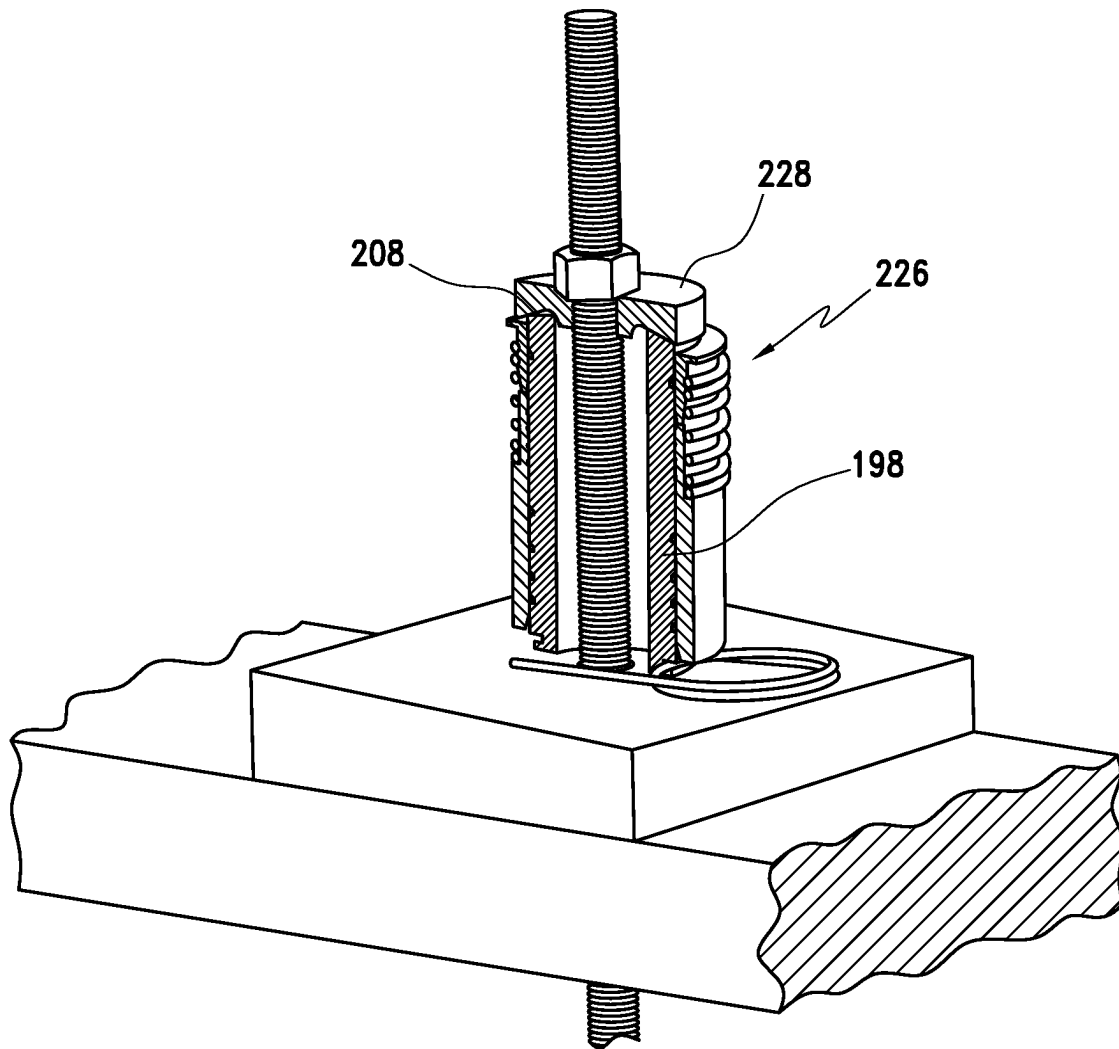


FIG. 33

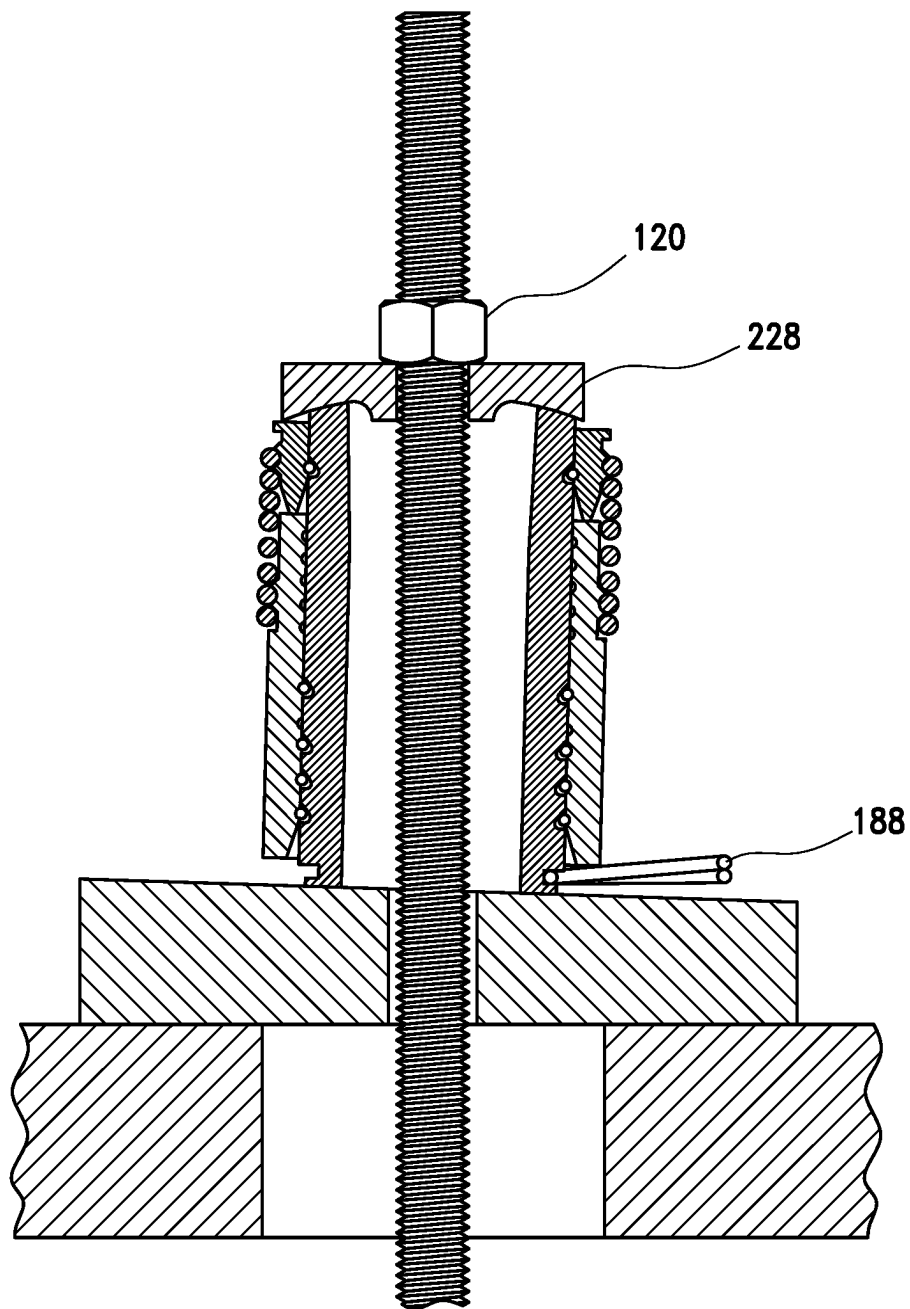


FIG. 34

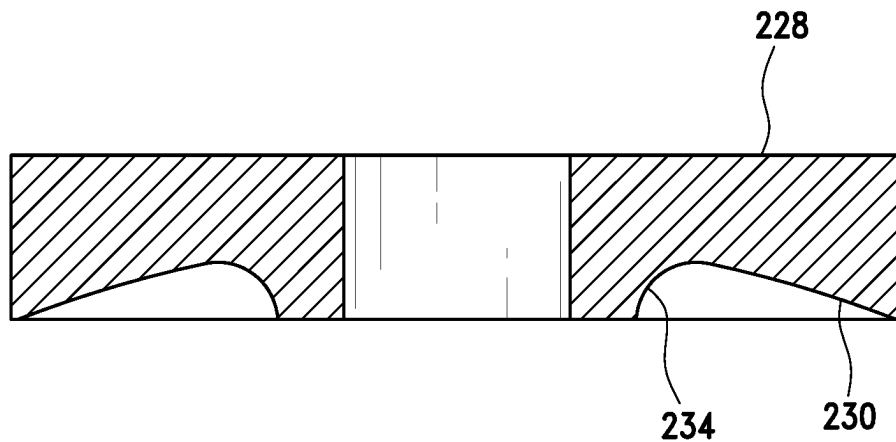


FIG. 35

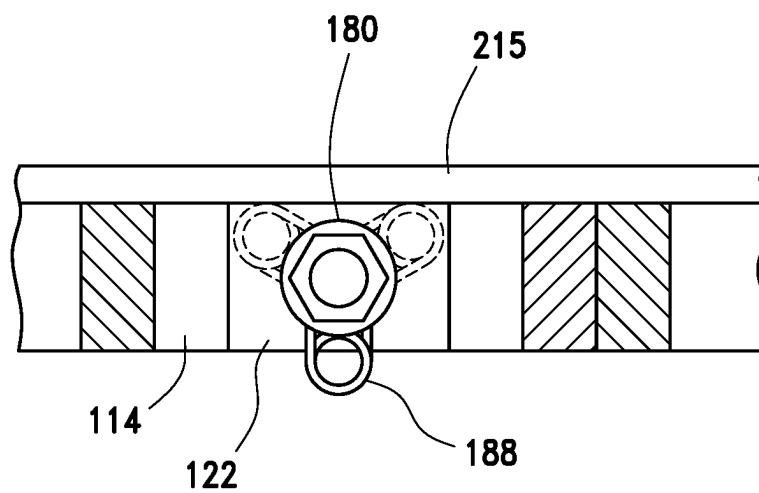


FIG. 36

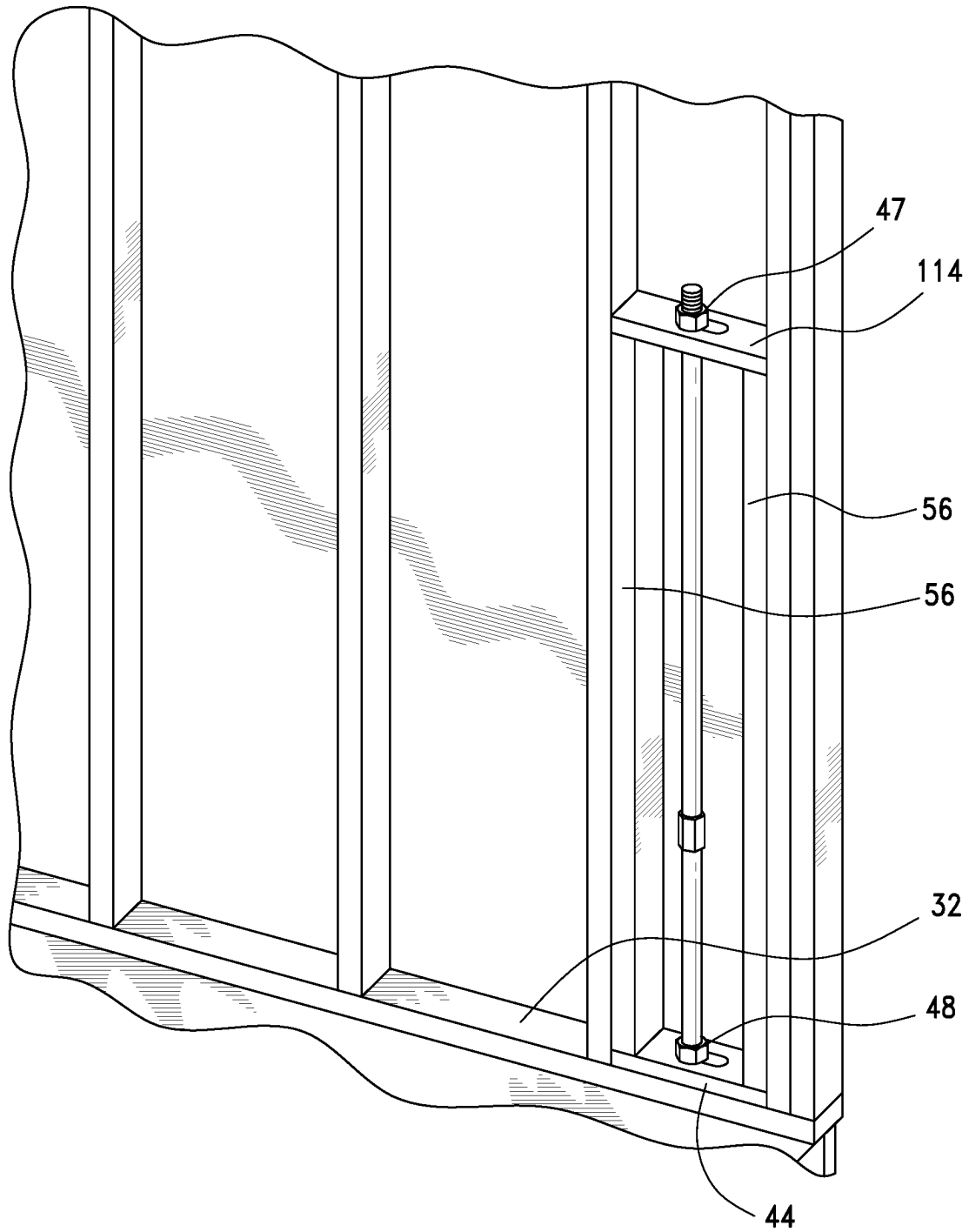


FIG. 37

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HOLD DOWN SYSTEM AND BUILDING USING THE SAME

RELATED APPLICATION

This is a continuation of application Ser. No. 15/041,786, filed Feb. 11, 2016, which is a continuation of International PCT Application No. PCT/US2007/019755, filed Sep. 12, 2007, which claims the priority benefit of provisional application Ser. No. 60/843,698, filed Sep. 12, 2006, all of which are hereby incorporated by reference.

FIELD OF THE INVENTION

The present invention is generally directed to a tension hold down system used in walls in light frame construction buildings to resist uplift and to compensate for wood shrinkage in wood frame construction and compression loading.

BACKGROUND OF THE INVENTION

Prior art hold down systems, such as one disclosed in U.S. Pat. No. 6,951,078, typically use a tie-rod that extends inside a stud wall from the foundation to the top floor. Field conditions and the quality of the installer are such that the tie-rod may not be installed perpendicular to the cross or bridge member to which the tie-rod is to be secured. In such a situation, the bearing surface of the bridge member and the hold down device may not align properly for good bearing contact due to the out-of vertical position of the tie-rod. The resulting installation may not be satisfactory since full contact between bearing surfaces of the various components of the system is not achieved, thereby compromising the integrity of the system.

Another problem encountered with prior art hold down devices is that sometimes the installer forgets to activate the device after installation. The problem is sometimes discovered after the wall has been closed, thus causing a lot of expense to correct the problem.

Yet another problem with prior art device is the need of using a tool to load the spring in a hold down device. When a hold down device comes preloaded from the factory, and an installer accidentally releases the lock and allows the compressed spring inside to expand, to reset the spring would typically require a tool which may not be available to the installer.

OBJECTS OF THE INVENTION

It is an object of the present invention to provide a hold down system and components therefor that would compensate for the non-perpendicularity of the tie-rod to the bearing surfaces of the wall to which the tie-rod is attached.

It is another object of the present invention to provide a hold down system and components therefor that would minimize the problem of forgetting to activate the hold down device after installation.

It yet another object of the present invention to provide a hold down system and components therefor that does not require the use of a tool to activate the hold down device.

It is still another object of the present invention to provide a hold down system and components where the hold down device can be reassembled and pre-loaded in the field by hand without using tools.

It is an object of the present invention to provide a hold down system and components therefor that is color coded to

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assist in the installation of the components that have been properly specified for the project.

In summary, a fastener assembly comprises a first cylindrical member disposed within a second cylindrical member, the first cylindrical member having an axial opening. One of the first and second cylindrical members is movable relative to another one of the first and second cylindrical members in a first direction, and the one of the first and second cylindrical members is locked relative to the another one of the first and second cylindrical members in a second direction opposite to the first direction. The first and second cylindrical members include first and second opposing cylindrical walls, respectively, the first opposing cylindrical wall including a plurality of first receiving volumes, and the second opposing cylindrical wall including a plurality of second receiving volumes. A spring is operably attached to the first and second cylindrical members to urge the one of the first and second cylindrical members in the first direction. A removable member is operably associated with the first and second cylindrical members to prevent the one of the first and second cylindrical members from moving in the first direction prior to removing the removable member, the removable member being configured to be pulled out to allow the one of the first and second cylindrical members to move in the first direction.

A fastener system for securing a building wall to a foundation comprises a tie-rod having one end for being secured to a building foundation and a building wall; a bearing plate for attachment to the wall, the bearing plate having an opening for allowing the tie rod to extend there-through; a first washer around the tie rod, the first washer having a first curved upper surface, the first washer for bearing on the bearing plate; a second washer around the tie rod, the second washer having a second curved bottom surface complementary to the first curved upper surface, the second curved bottom surface bearing on the first curved upper surface; and a nut for being secured against the second washer, the nut for tensioning the tie rod; whereby the first curved upper surface remains in substantial contact with the second curved bottom surface even when the tie rod is off vertical.

A fastener assembly comprises a first cylindrical member disposed within a second cylindrical member, the first cylindrical member having an axial opening. The second cylindrical member is movable relative to the first cylindrical member in a first direction, and the second cylindrical member is locked relative to the first cylindrical member in a second direction opposite to the first direction. A sleeve is attached to an upper end portion of the first cylindrical member; and a spring is operably attached to the second cylindrical member and the sleeve to urge the second cylindrical member in the first direction.

A fastener assembly comprises a first cylindrical member disposed within a second cylindrical member, the first cylindrical member having an axial opening for a tie rod. One of the first and second cylindrical members is movable relative to another one of the first and second cylindrical members in a first direction, and the one of the first and second cylindrical members is locked relative to the another one of the first and second cylindrical members in a second direction opposite to the first direction. A sleeve is disposed around the first cylindrical member, the sleeve being locked relative to the first cylindrical member in the first direction; and a spring is operably attached to the second cylindrical member and the sleeve to urge the one of the first and second cylindrical members in the first direction.

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a building comprises a stud wall including a first bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the first bottom plate within the stud wall; a first bridge member attached to the stud wall, the first bridge member having an opening through which the tie rod extends; a first fastener assembly connecting the tie rod and the first bridge member to the stud wall, the fastener assembly including an axial opening through which the tie rod extends. The first fastener assembly includes an inner body disposed within an outer body, a first nut securing the inner and outer bodies to the tie rod, a first spring urging one of the inner and outer bodies to move from a first position to a second position, and a removable restraining member locking the inner and outer bodies together to prevent the one of the inner and outer bodies from moving from the first position to the second position. The restraining member includes a handle portion extending from the outer body. The inner and outer bodies are movable relative to each other from the first position to the second position when the restraining member is pulled by the handle portion. The one of the inner and outer bodies is fixed relative to the tie rod and the another of the inner and outer bodies is fixed relative to the first bridge member such that when the tie rod moves through the stud wall, the one of the inner and outer bodies moves with the tie rod from the first position to the second position.

A building comprises a stud wall including a bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the bottom plate within the stud wall; a bridge member attached to the stud wall, the bridge member having an opening through which the tie rod extends; and a fastener assembly connecting the tie rod and the bridge member to the stud wall, the fastener assembly including an axial opening through which the tie rod extends. The fastener assembly includes an inner body disposed within an outer body, a first nut securing the inner and outer bodies to the tie rod, and a first spring urging one of the inner and outer bodies to move from a first position to a second position. The one of the inner and outer bodies is fixed relative to the tie rod and the another of the inner and outer bodies is fixed relative to the bridge member such that when the tie rod moves through the stud wall, the one of the inner and outer bodies moves with the tie rod from the first position to the second position. The first spring is configured to carry a load at least substantially equal to the weight of the length of the tie rod disposed between the fastener assembly and the foundation.

A building comprises a stud wall including a first bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the first bottom plate within the stud wall; a first bridge member attached to the stud wall, the first bridge member having an opening through which the tie rod extends; and a first fastener assembly connecting the tie rod and the first bridge member to the stud wall, the fastener assembly including an axial opening through which the tie rod extends. The first fastener assembly includes an inner body disposed within an outer body, a first nut securing the inner and outer bodies to the tie rod, and a first spring urging one of the inner and outer bodies to move from a first position to a second position. The one of the inner and outer bodies is fixed relative to the tie rod and the another of the inner and outer bodies is fixed relative to the first bearing plate such that when the tie rod moves through the stud wall, the one of the inner and outer bodies moves with the tie rod from the first position to the second position. A second bridge member is attached to the stud wall, the second bridge member having an opening

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through which the tie rod extends, the second bridge being disposed above and spaced from the first bridge member. A second fastener assembly is disposed above the first fastener assembly. The second fastener assembly includes a second inner body disposed within a second outer body, a second nut securing the second inner and outer bodies to the tie rod, and a second spring urging one of the second inner and outer bodies to move from a first position to a second position. The second spring is configured to carry a load at least substantially equal to the weight of the length of the tie rod disposed between the first and second fastener assemblies.

A building comprises a stud wall including a bottom plate and sheathing; a foundation; a tie rod operably anchored in the foundation and extending through the bottom plate within the stud wall; a bridge member having an opening through which the tie rod extends; and a fastener assembly connecting the tie rod and the bridge member to the stud wall, the fastener assembly including an axial opening through which the tie rod extends. The fastener assembly includes an inner body disposed within an outer body, a first nut securing the inner and outer bodies to the tie rod, a spring urging one of the inner and outer bodies to move from a first position to a second position, and a removable restraining member locking the inner and outer bodies together to prevent the one of the inner and outer bodies from moving from the first position to the second position. The restraining member includes a handle portion extending beyond the outer body such that the handle portion is prevented from being disposed between the outer body and the sheathing, thereby maintaining the handle portion in view of an installer. The inner and outer bodies are movable relative to each other from the first position to the second position when the restraining member is removed. The one of the inner and outer bodies is fixed relative to the tie rod and the another of the inner and outer bodies is fixed relative to the bridge member such that when the tie rod moves through the stud wall, the one of the inner and outer bodies moves with the tie rod from the first position to the second position.

A building comprises a stud wall including a first bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the first bottom plate within the stud wall; a first metallic bridge member disposed on the first bottom plate, the first metallic bridge member having an opening through which the tie rod extends; a first nut connecting the tie rod and the first bridge member to the first bottom plate; and a pair of spaced apart reinforcement studs. The pair of spaced apart reinforcement studs includes bottom ends bearing on the first metallic bridge member.

A building comprises a stud wall including a bottom plate; a foundation; a tie rod operably anchored in the foundation and extending through the bottom plate within the stud wall; a metallic bridge member attached to the stud wall, the bridge member having an opening through which the tie rod extends; and a fastener assembly connecting the tie rod and the bridge member to the stud wall, the fastener assembly including an axial opening through which the tie rod extends. The fastener assembly includes an inner body disposed within an outer body, a nut securing the inner and outer bodies to the tie rod, and a spring urging one of the inner and outer bodies to move from a first position to a second position.

A building comprises a stud wall including a bottom plate and a top plate; a foundation; a tie rod operably anchored in the foundation and extending through the bottom plate within the stud wall; a pair of spaced apart reinforcement studs including upper ends; a metallic bridge member bear-

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ing on the upper ends, the metallic bridge member having an opening through which the tie rod extends; and a nut connecting the tie rod and first bridge member to the reinforcement studs.

These and other objects of the present invention will become apparent from the following detailed description.

BRIEF DESCRIPTIONS OF THE DRAWINGS

FIG. 1 is a perspective view of stud wall of a one story building incorporating a hold down system made in accordance with the present invention.

FIG. 2 is a stud wall of a two story building incorporating a hold down system made in accordance with the present invention.

FIG. 3 is a perspective view of a stud wall of a two story building, incorporating a hold down system made in accordance with the present invention.

FIG. 4 is a perspective view of a stud wall using a hold down system made in accordance with the present invention.

FIG. 5 is a perspective view of a stud wall of a two story building utilizing a hold down system made in accordance with the present invention.

FIG. 6 is a perspective view of a stud wall of a two story building using a hold down system made in accordance with the present invention.

FIG. 7 is a perspective view of a stud wall of a three story building using a hold down system made in accordance with the present invention.

FIG. 8 is a perspective view of a bridge or cross member used in a hold down system shown in FIGS. 1-7.

FIG. 9 is a perspective view of an embodiment of a hold down device attached to a threaded rod.

FIG. 10 is an assembly view of FIG. 9.

FIG. 11 is a cross-sectional view of FIG. 9, showing the hold down device in a retracted position prior to extending to take up the shrinkage in the stud wall.

FIG. 12 is a cross-sectional view, showing the hold down device in a partly extended position after taking up the shrinkage in the stud wall.

FIG. 13 is an enlarged cross-sectional view taken from FIG. 12 of the cooperating grooves in the inner cylinder and outer sleeve or cylindrical member of the hold down device.

FIG. 14 is a perspective view of another embodiment of a hold down system made in accordance with the present invention.

FIG. 15 is an assembly view of FIG. 14.

FIG. 16 is a cross-sectional view of FIG. 14, showing the hold down device in a retracted position, prior to taking up the shrinkage in the stud wall.

FIG. 17 is a cross-sectional view of the hold down device of FIG. 14, shown in a partially extended position after taking up the shrinkage in the stud wall.

FIG. 18 is a perspective view of another embodiment of a hold down device made in accordance with the present invention.

FIG. 19 is an assembly view of FIG. 18.

FIG. 20 is a cross-sectional view of FIG. 18, showing the hold down device in a retracted position prior to taking up the shrinkage in the stud wall.

FIG. 21 is a cross-sectional view of FIG. 18, showing the hold down device in a partially extended position after taking up the shrinkage in the stud wall.

FIG. 22 is a perspective view of another embodiment of a hold down device.

FIG. 23 is an assembly view of FIG. 22.

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FIG. 24 is a cross-sectional view of FIG. 22, showing the hold down device in a retracted position prior to being activated after installation in the stud wall.

FIG. 25 is a cross-sectional view of FIG. 22, showing the hold down device in a partly extended position after having been activated and taken up the shrinkage in the stud wall.

FIG. 26 is a perspective view of another embodiment of a hold down device made in accordance with the present invention.

FIG. 27 is an assembly view of FIG. 26.

FIG. 28 is a cross-sectional view of FIG. 26, showing the hold down device in an off-vertical position and prior to being activated.

FIG. 29 is a cross-sectional view of FIG. 26, showing the hold down device in a partly extended position after having been activated.

FIG. 30 is a perspective view of a hold down assembly made in accordance with the present invention.

FIG. 31 is a cross-sectional view of FIG. 30.

FIG. 32 is an enlarged view of a portion of FIG. 5.

FIG. 33 is a perspective view, partly in cross-section, of another embodiment of a hold down device made in accordance with the present invention.

FIG. 34 is side elevation view of FIG. 33.

FIG. 35 is a cross-sectional view of a washer used in the embodiment of FIG. 33.

FIG. 36 is a cross-sectional view taken along line 36-36 in FIG. 32, showing the amount the spring clip extends out from the hold down device and its various positions, shown in phantom lines, around the hold down device.

FIG. 37 is another embodiment of a hold down system made in accordance with the present invention, showing the use of metallic bridge members and nuts alone.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIG. 1, a hold down system 2 made in accordance with the present invention is disclosed. The system 2 includes a foundation anchor 4 operably attached to a foundation 6 of a building. The foundation anchor 4 includes a threaded rod 8 attached to another threaded or tie-rod rod 10 by means of a coupling 12. A bridge member 14 spans between two adjacent studs 16 and is supported by a pair of reinforcement studs 18. A bearing plate 20 sits on top of the bridge member 14. The threaded rod 10 extends through the bridge member 14 and the bearing plate 20 through respective openings. A hold down device 22 is secured between the bearing plate 20 and a nut 24. The hold down device 22, which will be described below, is an expanding fastener assembly used to take up any slack that may develop in the tie-rod 10 due to shrinkage in the building wall.

FIG. 1 discloses a hold down system as used in a one story structure. The reinforcement studs 18 terminate between the top plate 26 and the bottom plate 28.

It should be understood that building foundation is used to refer generally to any structure that is used to anchor or tie a building to the ground. Examples are foundation walls, horizontal beams connected to vertical beams driven or buried in the ground, or any substantial structure solidly anchored in the ground. Accordingly, a building foundation can be any structure that is capable of transferring the load of the building to the ground.

FIG. 2 is an application of the hold down system 2 in a two story building. The hold down device 22 is disposed

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within the stud wall of the second floor. Reinforcement studs 27 extending from the bottom plate 28 to the top plate 26 are provided in the wall below the reinforcement studs 18 installed in the second floor wall. The bottom ends of the reinforcement studs 18 rest on the bottom plate 32 of the second floor wall.

FIG. 3 shows another embodiment of a hold down system 30 as applied to a two story building. The components are the same as the hold down system 2 shown in FIG. 1, except that a second hold down device 22 is disposed near the bottom plate 32 of the second floor wall. A second bridge member 14 bears directly on the bottom plate 32. A second bearing plate 20 bears on top of the second bridge member 14. The bottom ends of the reinforcement studs 18 bear down on the top surface of the second bridge member 14.

Another embodiment of a hold down system 34 is disclosed in FIG. 4 in a one story application. The system 34 is similar to the system disclosed in FIG. 1, except that a second bridge member 14 is disposed on the bottom plate 28 and secured by a nut 36 holding the second bridge member 14 tight against the bottom plate 28. The second bridge member 14 advantageously provides a bearing surface against the bottom plate 28 for distribution of forces that may tend to lift the wall off the foundation. The bottom ends 38 of the reinforcement studs 18 bear on top of the second bridge member 14.

Another embodiment of a hold down system 40 is disclosed in FIG. 5 in a two story application. The system 40 is similar to the system disclosed in FIG. 2, except that additional bridge members 42 and 44 are used. The bridge member 42 bears on top of the bottom plate 28 of the first floor wall, while the bridge member 44 bears on top of the bottom plate 32 of the second floor wall. A nut 46 secures the bridge member 42 to the tie rod 10 and helps secure the bottom plate 28 to the foundation. Similarly a nut 48 secures the bridge member 44 to the bottom plate 32 and to the tie-rod 10. The bottom ends 50 of the reinforcement studs 52 bear on top of the bridge member 42. The bottom ends 54 of the reinforcement studs 56 similarly bear on top of the bridge member 44. The reinforcement studs 52 extends the full height of the first floor wall, while the reinforcement studs 56 are shorter than the height of the second floor wall.

FIG. 6 discloses another embodiment of a hold down system 58 used in a two story building. The system 58 is similar to the system disclosed in FIG. 3, except that an additional bridge member 60 is provided that bears down on top of the bottom plate 28 and a nut 62 secures the bridge member 60 to the threaded rod 8. The ends of the reinforcement studs 52 bear down on top of the bridge member 60.

Another embodiment of a hold down system 63 applied to a three story building is disclosed in FIG. 7. The tie-rod 10 extends inside the stud wall through the first floor wall, second floor wall and terminates in the third floor wall. Bridge members 62, 64, 66, 68 and 70 along with hold down devices 72, 74, 76, 78 and 80 keep the tie rod 10 under tension. The bridge member 62 is supported by reinforcement studs 82 and 84 with the adjacent ends of the reinforcement studs sandwiching the respective ends of the bridge member 62. The hold down device 62 is disposed between a bearing plate 86 and a nut 88.

The hold down device 74 also bears down on a bearing plate 90 supported by the bridge member 64, which in turn bears down on the bottom plate. A nut 94 secures the hold down device 74 to the tie rod 10. The bottom ends of the reinforcement studs 100 bear down on the bridge member 64, transferring the load to the bottom plate and to the reinforcement studs 84 and 82 below.

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The hold down device 76 along with its bridge member 66 and a bearing plate 96 and its respective nut 98 are similarly installed as the hold down device 62. The reinforcement studs 100 and 102 similarly secure the bridge member 66 to the stud wall. The reinforcement studs 102 bear down on the bridge member 66, transferring the load to the reinforcement studs 100.

The hold down device 78 along with its bridge member 68, its bearing plate 104 and nut 106 are similarly secured as the hold down device 74. The ends of reinforcement studs 108 bear down on the bridge member 68, transferring the load to the base plate and to the reinforcement studs 102 below.

The bridge member 70 is supported on the top edge of the reinforcement studs 108 and is secured to the tie rod with nut 110. Bearing plate 112 is disposed between the bridge member 70 and the hold down device 80.

The various hold down systems disclosed above are shown installed within the first stud bay from the end of a shear wall using standard wood framing construction. However, the hold down systems are not limited to these locations or type of construction. They may be installed in any stud wall construction to resist uplift during high wind or earthquake conditions. The hold down system may be installed in the first stud bay at the first bay after a window or door opening. Generally, the hold down system may be installed anywhere inside a stud wall as the application dictates.

Referring to FIG. 8, a bridge member 114 used in the various embodiments of the hold down system of FIGS. 1-7 is disclosed. The bridge member 114 is a rectangular metallic block having a slotted opening 116 through which the threaded rod 10 extends. The opening 116 is centrally located and advantageously allows the threaded rod 10 to be slightly off center and off vertical when installed through the stud wall. The bridge member 114 is preferably made of aluminum, but other materials such as steel or non-metal materials may also be used. The metallic bridge member 114 simplifies the installation of a hold down system, requiring less number of components as compared to using a wood bridge typically made of several wood members. The metal bridge member 114 advantageously provides for higher loads as compared to wood bridge members, since "parallel to grain of lumber" loading is used (typically 1200 psi), as compared to "perpendicular to grain of lumber" loading when using wood bridge members (typically 625 psi). Referring to FIG. 7, the use of metallic bridge members 62 and 66 where the reinforcement studs 84 and 102 bear down from above advantageously eliminates the "perpendicular to grain" loading of prior art wood bridge member, thereby increasing the loading capacity of the hold down system. The bridge member 114 may be color coded for material type, capacity and dimensional size.

The bridge member 114 is also used as a baseplate compression plate as shown in FIGS. 3, 4, 5, 6 and 7. As load passes through the support studs and or wall studs through the parallel wood grain, this surface is in bearing contact with each end of the metallic baseplate. The use of the bridge member as a base-plate-compression plate lowers the compression force per square inch upon the perpendicular wood bearing surface below. As load is transferred from the support studs and or wall studs through the metallic baseplate, the load is dispersed and spread out because the metallic baseplate is minimally designed not to bend or deflect. The physical properties of the metallic baseplate provide this behavior when used in this fashion. So, a concentrated force from the contact point of the studs at each

end of the top the metallic baseplate is then spread out over the large area of contact to the perpendicular wood bearing surface underneath the metallic baseplate.

Placement of the metallic baseplate and bridge member is intended for the relative center of the first stud bay of a wall in a building which uses wall studs of many different types of framing material. They may also be installed at each end of a wall. They may also centrally be located in any stud bay of a wall or every stud bay of a wall. The transfer of parallel to grain force or load from support studs and or wall studs bearing upon the upper top side of the metallic bridge block is transferred to the lower support studs and or wall studs through the metallic bridge member. The metallic bridge member physical properties do not allow any crushing or displacement between studs parallel to grain bearing surfaces; therefore, force or load is transferred with a stable load path.

Bridge member or baseplate compression plate is not limited to metallic materials. The physical properties of the cross member and or baseplate compression plate must be equal to or greater than the physical properties of the support studs bearing surface.

Bridge member and/or baseplate compression plate can be employed to resist uplift and rotation of a wall of a building and also are utilized when the wall in a compression mode. Because of behaviors described earlier above the bridge member and/or baseplate compression plate disperses loads and achieves lowering concentrated forces between bearing surfaces when down-load forces occurs. This advantageously helps solve load path problems in current hold down systems.

Various embodiments of a hold down device as used in the hold down systems of FIGS. 1-7 will now be described. A person of ordinary skill in the art will appreciate that the various components disclosed for each hold down device may be interchanged, substituted, added, or deleted as desired by the application.

A hold down device **118** made in accordance with the present invention is disclosed in FIGS. 9-13. The hold down device **118** is shown attached to the tie rod **10** by means of a nut **120**, bearing plate **122** and the bridge member **114**. The hold down device **118** includes an inner cylinder **124** slidably disposed within an outer sleeve or cylindrical member **126**. A top cylinder or sleeve **128** is secured to an upper portion of the inner cylinder **124**. A coil spring is compressed between the top cylinder **128** and the outer sleeve **126**. A convex washer is disposed between the nut **120** and the inner cylinder **124**. The washer **132** has a convex surface **134** which matches a corresponding concave surface on the upper edge portion of the inner cylinder **124**. The inner cylinder has central opening **138** which is larger than the diameter of the threaded rod **10** to allow the rod to be slightly off vertical within the inner cylinder **124**. The opening **138** also flares out to a larger diameter at the top portion of the inner cylinder **124** (see FIGS. 11 and 12 and FIGS. 28 and 29) to allow even further leeway for the tie-rod to be off perpendicular with respect to the bridge member **114**. The convex surface **134** and the concave surface **136** are advantageously in full bearing contact when the threaded rod **10** is slightly off vertical.

The top cylinder **128** is secured to the inner cylinder **124** by means of a C-ring or resilient member **140** captured in a circumferential groove or receiving volume **142** around the outside cylindrical surface of the inner cylinder **124** and an inner circumferential groove or receiving volume **144** along the inner cylindrical surface of the top cylinder **128**. The grooves **142** and **144** are configured with the same geometry

as disclosed in U.S. Pat. No. 6,951,078, hereby incorporated by reference. The geometry of the grooves **142** and **144** is such that the top cylinder **128** is prevented from sliding upwardly with respect to the inner cylinder **124** but is able to slide downwardly for assembly purposes. The lower portion of the inner cylinder **124** includes a plurality of circumferential grooves or receiving volumes **146** with the same geometry as the circumferential groove **142**. Similarly, the inner cylindrical surface of the outer sleeve **126** has a plurality of circumferential grooves or receiving volumes **148** that has the same geometry as the circumferential groove **144**. A plurality of C-rings or resilient members **150** are disposed around the grooves **146**. The grooves **146** and **148** allow the vertical sliding movement of the inner cylinder **124** with respect to the outer sleeve **126**. An annular groove **151** is disposed adjacent the bottom edge of the inner cylinder **124** for use with a removable spring clip **188** (see, for example, FIG. 24) prior to installation. The groove **151** is present in the other embodiments shown in FIGS. 14-21.

Details of the grooves **146** and **148** are disclosed in FIG. 13. Downward movement of the outer sleeve **126** will cause the C-ring **150** into the groove **146**, which is sized to completely receive the cross-sectional area of the C-ring, thereby allowing further movement of the outer sleeve **126**. On the other hand, upward movement of the outer sleeve **126** is prevented, since the groove **148** is not large enough to accommodate the entire cross-sectional area of the C-ring **150**.

In operation, the spring **130** is under compression, as shown in FIG. 11. The threaded rod **10** is under tension from the action of spring **130**, which tends to pull the threaded rod **10** upwardly. As the building wall shrinks over time, the bridge member **114** would move downwardly with the shrinkage of the wall. This causes the spring **130** to push the outer sleeve **126** downwardly while at the same time pushing the top cylinder **128** upwardly, thus causing the inner cylinder **124** to move upwardly. With these relative motions, tension on the threaded rod **10** is thus maintained. Once the hold down device **118** has expanded as shown in FIG. 12, it is prevented from contracting back to its original position as shown in FIG. 11 by virtue of the geometry of the cooperating grooves **146** and **148** as explained above.

The top cylinder **128** has outer helical (thread-like) grooves **152** for receiving an end portion of the spring **130**. Similarly, the outer sleeve **126** has outer helical (thread-like) grooves **154** for receiving the opposite end portion of the spring **130**. The grooves **152** and **154** keep the top cylinder **128** and outer sleeve **126** attached to each other via the spring **130**.

Another embodiment of a hold down device **156** is disclosed in FIGS. 14-17. The hold down device **156** is identical to the hold down device **118** except for the provisions of a concave washer **158** welded to the bearing plate **122** and the provision of a convex outer edge surface **159** on the outer cylinder **160**. The hold down device **156** advantageously allows for a greater deviation from the vertical for the threaded rod **10** while still maintaining full bearing contact between the outer cylinder **160** and the concave washer **158** and between the convex washer **132** and the concave edge surface of the inner cylinder **136**, since adjustment at the convex washer **132** is independent of the adjustment at the concave washer **158**. FIG. 16 shows the hold down device **156** in a retracted position. FIG. 17 shows the hold down device **156** in an expanded position after the building wall has shrunk and the hold down device **156** has taken up the resulting slack in the threaded rod **10**, thereby keeping the threaded rod **10** in tension.

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Another embodiment of a hold down device **162** is disclosed in FIGS. **18-21**. The hold down device **162** is similar to the other disclosed hold down devices, except for some features. A nut **164** has a concave surface **166** that mates with the corresponding convex edge surface **168** on the inner cylinder **170**. The bearing plate **172** has a convex surface **174** that mates with a concave edge surface **176** on the outer sleeve or cylindrical member **178**. The bottom edge **177** of the inner cylinder **170** also includes a mating concave surface. All the other components of the hold down device **162** are identical to similar components in the previously described hold down devices. FIG. **20** shows the hold down device **162** in a retracted position while FIG. **21** shows the device in the extended position after taking up the shrinkage in the wall.

Another embodiment of a hold down device **180** is disclosed in FIGS. **22-25**. The hold down device **180** includes an inner cylinder **182** and outer sleeve or cylindrical member **184**, a top cylinder or sleeve **128**, a flat washer **186** and a nut **120**. As in the other embodiments, spring **130**, C-rings **140** and **150** are also provided. The inner cylinder **182** includes a greater number of circumferential grooves **146**, a corresponding number of circumferential grooves **148** on the inside cylindrical surface of the outer sleeve **184**, and a corresponding number of C-rings **150**. This is to advantageously provide for an increased load for the hold down device **180**. A spring clip **188** advantageously allows the hold down device **180** to be preloaded (when the spring **130** is compressed) prior to installation. The spring clip **188** has two extending legs **190** and a portion **192** that slip into an annular groove **194** so that the outer sleeve **184** is stopped from moving downwardly due to the action of the compressed spring **130**. Thus, the hold down device **180** can be installed with the spring **130** already loaded. After the hold down device **180** has been installed in place, as shown in FIG. **24**, the spring clip **188** is then pulled out, allowing the outer sleeve **184** to bear down on the bearing plate **122**, ready to take any slack on the threaded rod **10** due to any shrinkage in the wall, as shown in FIG. **25**.

The annular groove **194** includes a bottom wall **195** which is smaller in outside diameter than the diameter of the inner cylinder **182** to prevent any deformations or roughness, such as burrs, on the bottom wall **195** from interfering with the inner cylindrical surface of the outer sleeve **184** in case such deformations or roughness are inadvertently formed when the spring clip **188** is pulled out. This feature of the annular groove **194** is present in all of the embodiments disclosed herein.

Another embodiment of a hold down device **196** is disclosed in FIGS. **26-29**. The hold down device **196** includes an inner cylinder **198**, an outer sleeve or cylindrical member **200**, a top cylinder **128**, a concave washer **202**, a flat washer **204**, a nut **120**, C-rings **140** and **150**, a spring **130** and a spring clip **188**. The concave washer **202** has a concave surface **206** that mates with a convex surface **208** on the upper edge of the inner cylinder **198**. The spring clip **188** is removably retained within the groove **194** on the inner cylinder **198**. A bearing plate **210** has a ramp surface **212** and is used to compensate for the misalignment of the threaded rod **10** out of the vertical. The ramp surface **210** disclosed is about 2° from the horizontal, but other angles may be incorporated depending on need. Additional compensation for the misalignment of the threaded rod **10** from the vertical is provided by the concave washer **202**, as previously discussed in connection with the other embodiments. Prior to the spring clip **188** being pulled out to activate the hold down device **196**, a gap **214** is advantageously clearly

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visible between the bottom edge of the outer sleeve **200** and the top surface of the bearing plate **210**. The gap **214** may be used for quality control as an indicator on whether the hold down device has been activated or not. If the gap **214** exists, it means that the spring clip **188** has not yet been pulled out. The spring clip **188** extends out in such a distance that it cannot fit behind the hold down device and be hidden from view. Preferably, the spring clip **188** should stick out in front so that it will have to be pulled out before a wallboard can be installed, since it will be in the way of the wallboard. Referring to FIG. **36**, the spring clip **188** extends a distance from the hold down device **180** such that it cannot be placed inadvertently between the wall sheathing **215** and the hold down device **180**. In this manner, the spring clip **188** will always be visible, as shown in phantom lines, as a reminder to the installer that it needs to be pulled out before the wall is closed off. Note the larger and flared out opening at the top portion of the inner cylinder **198** that allows the tie-rod to be off-center with respect to the hold down device **196**.

An embodiment of a hold down assembly **216** is disclosed in FIGS. **30** and **31**. The hold down assembly **216** includes the threaded rod **10** attached to the stud wall by means of the bridge member **114**, bearing plate **122**, a nut **120**, concave washer **218** and convex washer **220**. The concave washer **218** has a concave surface **222** that mates with a corresponding convex surface **224** on the convex washer **220**. The washers **220** and **222** allow the threaded rod **10** to be out of the vertical while maintaining a maximum bearing contact with the bearing plate **142**. The washers **218** and **220** allow for centering the rod **10** while providing full bearing contact between bearing surfaces. The washers **218** and **220** may also be used in the embodiments shown in FIGS. **1-7** and **37** where a nut is used directly to hold down a bridge member without an intervening hold down device.

The convex and concave washers used in the various embodiments advantageously provide constant bearing area load transfer connections. The convex and concave washers advantageously provide for self-centering installation where the threaded rod is out of the vertical. The convex and concave washers may be color coded to indicate the design load, capacity, hole size and/or diameter. The color coding may also indicate the amount of rotation or swivel provided. The convex or concave washers may be used on top or underneath the hold down device. The convex or concave washers may be integrated or fabricated onto the surface of the bearing plate. The use of convex and concave washers and correspondingly shaped edge surfaces on the hold down device advantageously allow for swivel or rotation when connected to tension members such as a threaded rod inside a wall that is not perpendicular to the bearing surface of the wall, thereby providing a constant bearing area between bearing surfaces.

The spring clip **188** is also used in the other embodiments of the hold down device, shown in FIGS. **9-21**, although not shown therewith. Thus, the hold down devices disclosed herein are activated without use of tools.

The devices may also be reassembled and pre-loaded manually using one's hands and without the use of tools. Referring to FIGS. **23** and **24**, for example, the spring **130** is attached to the outer cylinder or sleeve **184** and the top cylinder or sleeve **128**. The C-rings **150** and **140** are placed within the circumferential grooves **146**. The C-rings **150** and **140** will be loose, since they are biased toward the circumferential grooves **148** (see FIG. **13**) on the inner surface of outer cylinder **184**. The inner cylinder **182** is then inserted into the outer cylinder **184** from below. The retaining member legs **190** are inserted into the annular groove **194**.

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The top cylinder or sleeve **128** and the spring **130** are then pressed down until the top cylinder **128** locks with the inner cylinder **124** via the C-ring **140**.

The spring clip **188** cannot be hidden from sight when placed within or on a 2×4 and 2×6 typical framed wall. The clip's own dimension or diameter will not allow it to be rotated out of sight if any type of wall sheathing or wall board is present. This safeguard is so that the clip will not be forgotten and the device rendered un-activated and useless.

The various embodiments of the hold device disclosed herein provides linear non-rotating or axial motion that slides over and around various common tension materials, such as the tie-rod **10**, with common hardware fastener connection. The hold down devices may be color coded to indicate travel or stroke length, capacity of useable force or load, and/or inside or outside diameter of the inner cylinder.

The spring **130** is designed and/or engineered to have a potential energy equal to or greater than the weight of the length of the tie-rod or cable (when used) below the hold down device extending to the next below hold down device or to the foundation anchor. The spring **130** may also be designed or engineered to have a minimum tension or force equal to or greater than the weight of the tie-rod or cable at full travel stroke length or designed displacement of the device. In this manner, the tie-rod or cable is advantageously fully supported throughout its length, regardless of the amount of travel of the of the outer sleeve relative to the inner cylinder, thereby preventing any buckling or bowing of the tie-rod or cable due to its own weight.

FIG. **32** is an enlarged view of a portion of FIG. **5** at the second floor. Note the nut **48** securing the bridge member **44**, used as baseplate compression plate, to the bottom plate and the tie-rod. Also note the bridge member **114** bearing down on the end grain of the reinforcement studs **56** in a direction parallel to the grain of the wood.

FIG. **37** is similar to FIG. **32** but showing the absence of the hold down device **180** and the bearing plate **122**. Nut **47** secures the bridge member **114** to the top ends of the reinforcement studs **56**, which are nailed to the adjacent studs. The washers **218** and **220** disclosed in FIG. **31** may also be used in conjunction with the nuts **47** and **48**.

Another embodiment of a hold down device **226** is disclosed in FIGS. **33-35**. The hold down device **226** is similar to the hold down device disclosed in FIG. **26**, except that a different washer **228** is provided. The washer **228** has a circumferential concave surface **230** that mates with corresponding convex edge surface on the inner cylinder **198**. The concave surface **230** terminates into a curved wall **234**. The curved wall **234** provides a centering function and limits the lateral adjustment of the washer **228** relative to the hold down device **226**, as shown in FIGS. **33** and **35**.

It should be understood that the use of a bearing plate in conjunction with the metallic bridge member disclosed in the various embodiments is optional. The metallic bridge member may be used without the bearing plate.

The hold down device disclosed herein may also be used as a tensioning device when used as an expanding washer or expanding sleeve that takes up slack that may develop in a bolt, cable, tie rod, etc. used in any structure requiring maintenance of applied tension.

While this invention has been described as having preferred design, it is understood that it is capable of further modification, uses and/or adaptations following in general the principle of the invention and including such departures from the present disclosure as come within known or customary practice in the art to which the invention pertains,

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and as may be applied to the essential features set forth, and fall within the scope of the invention.

I claim:

1. A building, comprising:

- a) a first stud wall including a first bottom plate disposed on top of a foundation;
- b) a threaded anchor rod operably anchored in the foundation and extending through the first bottom plate within the first stud wall;
- c) a first bridge member disposed on the first bottom plate, the first bridge member having a first opening, the threaded anchor rod including an end portion extending through and past the first bridge member;
- d) a coupling separate from the first bridge member;
- e) the coupling directly connecting the end portion of the threaded anchor rod to an end portion of a threaded tie rod, the threaded tie rod extending upwardly from the coupling in the first stud wall;
- f) a pair of spaced apart studs;
- g) the pair of spaced apart studs including bottom ends bearing on the first bridge member;
- h) a second bridge member having a second opening, the tie rod extending through and past the second opening, the second bridge being disposed above and spaced from the first bridge member;
- i) a fastener attached to the tie rod to bear down on the second bridge member; and
- j) the pair of spaced apart studs including top ends, the second bridge member bearing on the top ends; and
- k) a metal bearing plate disposed between the second bridge member and the fastener.

2. The building as in claim 1, wherein the fastener includes a nut.

3. The building as in claim 1, wherein the first bridge member is made of metal.

4. The building as in claim 1, wherein the second bridge member is made of metal.

5. The building as in claim 1, and further comprising a fastener threaded to the threaded anchor rod and bearing on the first bridge member.

6. The building as in claim 5, wherein the fastener includes a nut.

7. A building, comprising:

- a) a first stud wall including first and second vertical studs with respective first and second top ends;
- b) a foundation;
- c) a tie rod operably anchored in the foundation;
- d) a second stud wall disposed above the first stud wall;
- e) a first metallic bridge member disposed on and supported by the first and second top ends, the first bridge member having a first opening, the tie rod extending from the foundation through the first opening and past the first bridge member to the second stud wall; and
- f) a fastener attached to the tie rod to bear on top of the first bridge member to tension the tie rod.

8. The building as in claim 7, wherein:

- a) the second stud wall includes a second bottom plate;
- b) a second bridge member with a second opening is disposed on the second bottom plate; and
- c) the tie rod extends through the second opening past the second bridge member.

9. The building as in claim 8, wherein:

- a) the second stud wall includes third and fourth vertical studs with respective third and fourth top ends;
- b) a third bridge member with a third opening is disposed on the third and fourth top ends; and

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c) the tie rod extends through the third opening past the third bridge member.

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